

Research paper

Robust and efficient single-CNN-based spacecraft relative pose estimation from monocular images

Michele Bechini^{ID}*, Michèle Lavagna^{ID}

Politecnico di Milano, Dept. of Aerospace Science & Technology, via La Masa 34, 20156 Milano, Italy

ARTICLE INFO

Keywords:

Relative pose estimation
Vision-based navigation
Convolutional neural networks
Synthetic augmentations
Keypoints-based methods

ABSTRACT

Autonomous spacecraft relative navigation is crucial for future space missions, but achieving accurate and efficient pose estimation, especially for uncooperative targets, remains challenging. Despite recent strides in AI-based solutions and benchmark datasets, existing algorithms prioritize accuracy over computational efficiency and struggle to generalize from synthetic to real-world scenarios. The paper addresses these issues by proposing a novel pose estimation algorithm from monocular images based on a single multitasking convolutional neural network (CNN) performing both region of interest (ROI) estimation and keypoint regression. The proposed architecture leverages an outlier rejection scheme based on confidence scores to enhance robustness and reliability, while a check on the need for a second inference has been introduced to improve the computational efficiency and potentially enable higher navigation filter update rates. To improve the domain gap bridging capabilities of the introduced pipeline, custom augmentation techniques are presented within this work, including a novel noise augmentation mimicking actual sensor noise. These augmentations have been used during the model training and proved to be beneficial in enhancing the CNN performances, setting the new highest score for object detection and keypoint regression on a benchmark dataset. The performances of the proposed architecture have been assessed using synthetic SPEED images. The outcomes of the analyses demonstrate that our architecture achieves high-level performances on synthetic SPEED images with a mean translational error lower than 10 cm and a mean angular error of about 1.4 degrees, outperforming other more complex models while maintaining an execution time on the CPU of an Apple® Silicon™ M1 Pro processor in the order of 60 ms to 131 ms, depending if one or two inferences are needed to retrieve the pose. Further, the standard deviations of the errors are 11.5 cm and 1.0 degree for the translation and attitude errors, respectively, revealing the high precision of the proposed solution and the absence of strong outliers, especially for the relative attitude estimation where the registered standard deviation is the lowest among the methods available in the literature. The evaluation using mock-up SPEED+ frames confirms the effectiveness of the introduced domain gap reduction strategies, with performance metrics remaining competitive despite increased errors compared to synthetic images mainly due to low illumination conditions. Notably, the paper also outlines the extension of the proposed architecture to more complex targets to prove the adaptability of the proposed approach. The outcomes of this analysis confirm that the introduced pipeline still achieves high accuracy in the relative pose estimation tasks even for targets with complex geometries and a high probability of keypoints occlusions, as for the case of Envisat. Conversely, for highly symmetric targets like VESPA, the performances of the keypoint regression degrades, leading to wrong estimates due to the uncertainties in the retrieved keypoints locations.

1. Introduction

Autonomous spacecraft relative navigation stands as a pivotal technology for forthcoming space missions where the accomplishment of objectives in highly challenging operational scenarios necessarily entails a high degree of autonomy of the involved spacecraft [1]. Autonomous navigation capabilities are crucial for proximity operations,

encompassing tasks like active debris removal and on-orbit servicing [2–4], spacecraft rendezvous [5], docking [6], precision landing maneuvers on major celestial bodies [7–9] and small asteroids [10], as well as for rovers moving in an unknown environment [11]. Similarly, in formation flying missions, maintaining designated formations [12] or executing collision avoidance maneuvers when operating in proximity

* Corresponding author.

E-mail addresses: michele.bechini@polimi.it (M. Bechini), michelle.lavagna@polimi.it (M. Lavagna).

<https://doi.org/10.1016/j.actaastro.2025.04.016>

Received 2 October 2024; Received in revised form 5 April 2025; Accepted 8 April 2025

Available online 17 April 2025

0094-5765/© 2025 IAA. Published by Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

of space-resident objects [13] necessitates robust relative navigation solutions. Notably, the best performances are achieved by relying on onboard relative state estimation algorithms to ensure responsiveness and efficacy in both nominal and off-nominal conditions. Usually, a relative navigation pipeline comprises two steps, i.e., the relative state acquisition (from a single frame or concatenation of frames [14]) and the subsequent filtering phase to smooth possible errors and outliers [15,16]. Being these two steps concatenated, performing an accurate relative pose estimation entails a reduction of the error in the subsequent filtering phase and, as a consequence, of the overall relative navigation error, improving the performances of the entire guidance navigation and control (GNC) chain. Target-chaser systems represent one of the most attractive and challenging scenarios for relative pose estimation tasks and, within this framework, the case of uncooperative targets and no a priori knowledge of the relative state is of particular interest [17]. In these conditions, the relative state shall be estimated autonomously by relying only on the capabilities and sensors of the chaser. Monocular cameras are among the most adopted sensors to retrieve the relative pose from the acquired images [17,18]. Notably, on top of the possibility of relying on images acquired at different spectral bands such as the visible (VIS) range, the most widely adopted, and the thermal-infrared (TIR) range as in [19,20], monocular cameras are more affordable in terms of mass and power requirements than light detection and ranging (LiDAR) and more complex sensors [14], making them suitable for a wider range of missions.

Techniques to estimate the relative pose from monocular images are well documented in the literature, and, in the last few years, new solutions based on artificial intelligence (AI), mostly relying on convolutional neural networks (CNNs), have been developed and proven to achieve new state-of-the-art performances [21,22], thanks to their effectiveness in automatically learning hierarchical image representations. The development of AI-based solutions was pushed further by the release of two benchmark datasets, SPEED [23] and SPEED+ [24] (both using Tango from PRISMA mission [25] as target), and the related Satellite Pose Estimation Challenges held in 2019 (SPEC2019) [26] and 2021 (SPEC2021) [27]. The outcomes of both the challenges prove that entrusting the target detection and feature regression tasks to dedicated CNNs and then relying on classic solvers for the Perspective- n -Point (PnP) problem (e.g., the EPnP solver [28]) to retrieve the final relative pose is the most effective approach [26]. Despite that, it has also been noticed in [29] that top-level performances were achieved by computationally expensive architectures that maximize the accuracy of the estimates without taking into account the computational footprint (mainly related to the adopted CNNs), resulting in algorithms with limited possibilities to be deployed onboard, due to the constrained memory and computational resources offered by the currently available on-board computers [29,30]. Additionally, the limited capabilities of bridging the domain gap between synthetic training images and the actual mock-up frames used in the testing phase [31] for most of the architectures proposed was further assessed in [27], posing a new challenge for the applicability of most of the algorithms in real case scenarios. While this work focuses on domain gap bridging techniques, it is worth mentioning that domain adaptation strategies, such as unsupervised domain adaptation, can complement these approaches. An example is the transformer-based UDFormer introduced in [32], the first transformer-based unsupervised domain adaptation framework. UDFormer is designed for Mars landscape scenes and employs a teacher–student model where the pseudo-labels predicted by the teacher are used to supervise the training of the student model, which is performed by also leveraging the true labels of the source domain. Its self-training approach, aided by stochastic data augmentation and output-guided biased sampling, can inspire similar methods for spacecraft pose estimation, helping address domain shifts in lighting, noise, and other variations between synthetic and real images.

Thus, the current limitations of most relative pose estimation algorithms based on CNNs coupled with PnP solvers comprise:

- **High Computational Demands:** Most state-of-the-art solutions rely on complex architectures that require significant computational resources, hindering their deployability on resource-constrained onboard computers.
- **Domain Gap Challenges:** A substantial performance gap exists between synthetic training data and real-world images, limiting the practical applicability of many algorithms that does not generalize well.
- **Limited Generalization:** Existing methods are typically tailored to specific scenarios, such as those of the SPEED/SPEED+ datasets, reducing their adaptability to diverse target geometries and operational conditions.

Relying on the capabilities of lightweight CNNs in relative pose estimation tasks demonstrated in [33], a novel, accurate, efficient, and robust relative pose estimation algorithm based on a single CNN is proposed here. Namely, being inspired by the more recent developments in the field of human pose estimation (HPE), the algorithm proposed here relies on a CNN trained to learn the specific position of each keypoint in each image, as in [21], to reduce the post-processing time for the regressed features and make the adaptation of the proposed pipeline to different target geometries easier than the case of algorithms leveraging other features (e.g., the line segments adopted in [33,34]). The single CNN leveraged within this work is the YOLOv8-pose¹ model, being capable of accurately performing target detection and keypoints regression through a single inference despite being among the lightest keypoint regression CNN models [29,35]. The proposed architecture is more efficient than the one in [33] since it solves only a PnP by leveraging the single 2D-to-3D correspondence retrieved, achieving a lower overall running time on CPU, despite relying on a CNN model slightly more complex than the lightweight ones adopted in [33]. Among the main novelties, the proposed architecture also relies on a check on the actual need for a second inference on the ROI-cropped image that is performed based on the information extracted by coupling the ROI size and the confidence associated with keypoints after the first inference, further improving the efficiency of the entire pipeline and reducing the overall running time in the most critical phases of target inspection (i.e., the closest approach). Further, an outlier rejection step based on the confidence score associated with each keypoint enhances the robustness and accuracy of the pipeline. Lastly, despite the work in [33] already demonstrated that lightweight CNNs are capable of partially bridging the domain gap between synthetic and actual mock-up images, custom synthetic image augmentation strategies, including a novel high-fidelity VIS sensor noise generator, are introduced within this work and adopted during the training phases of the involved CNN, proving their effectiveness in boosting both the performances on synthetic images and actual mock-up frames. As principal outcomes, the proposed architecture, trained using the aforementioned synthetic augmentations, achieves the new highest score for target detection and keypoint regression on synthetic SPEED images. Further, the accuracy and precision of the relative pose estimate are comparable with those of most complex architecture both for SPEED synthetic images and SPEED+ mock-up frame, proving the effectiveness of the proposed approach. The main contributions introduced within this work are as follows:

- A lightweight, robust, and efficient relative pose estimation algorithm is proposed, leveraging the YOLOv8s-pose for ROI and 2D keypoint estimation, with novel inference-management and outlier rejection strategies to enhance accuracy and efficiency during close target approaches.
- Synthetic image augmentations, including a novel high-fidelity VIS sensor noise generator, are introduced to improve the generalization of the model across domains, boosting the performance of YOLOv8s-pose on both synthetic and real mock-up images.

¹ <https://github.com/ultralytics/ultralytics>.

- Experimental results on Tango (SPEED/SPEED+), Envisat, and VESPA demonstrate the effectiveness and adaptability of the proposed pipeline to non-symmetric target geometries and highlight its limitations for highly symmetric targets.

In the remainder of the paper, the works related to the autonomous relative pose estimation and keypoints regression via AI-based methods are reviewed in Section 2. The proposed architecture is detailed in Section 3, providing a comprehensive overview of all its main processing steps, as well as insights on the YOLOv8-pose model and the new augmentation strategies introduced. The results achieved with the YOLOv8s-pose after the training phase and the performance assessments of the overall architecture are discussed in Section 4. Namely, after introducing the evaluation metrics in Sections 4.1, 4.2 outlines the YOLOv8s-pose training process and the ablation study performed to select the best model, also presenting the main achievements in terms of target detection and keypoints regression with respect to other models available in the literature. Then, Sections 4.3.1 and 4.3.2 report the outcomes of the proposed architecture on images from SPEED and SPEED+, respectively, providing also a comparison against other algorithms that participated in SPEC2019 and SPEC2021. Further, Section 4.4 provides the outcomes of the analyses performed after adapting and testing the proposed architecture on images containing Envisat and VESPA as targets to assess the performances for more complex target geometries. Lastly, the conclusions and possible future developments are reported in Section 5.

2. Related works

The relative pose initialization consists of retrieving the relative state of a target without any a priori knowledge, and it is suitable for the first estimate (i.e., the initialization) of the relative state. For the subsequent steps, it is possible to rely either on a new initialization for each frame or on pose-tracking algorithms to evaluate the evolution of the initial relative pose through consecutive frames [14]. Relative pose estimation algorithms are usually classified into model-based [17], non-model-based [36–38], and hybrid [18] approaches. The model-based approaches are the most well-suited for the case of known uncooperative targets since they rely on the knowledge of the 3D target geometry [14]. Namely, the frames acquired by the camera of the chaser are processed via dedicated image processing (IP) algorithms that retrieve the 2D features. Then, these features are matched with the corresponding ones retrieved from the known 3D model and fed to a PnP solver [28,39] that outputs the relative pose that best approximates the reprojection of the 3D features into the 2D ones. This relative pose estimate can be further refined using numerical optimization methods such as Levenberg–Marquardt [40] or Newton–Raphson [41]. Consequently, the correctness and accuracy of the extracted 2D keypoints and their matching with those from the known 3D model are fundamental in achieving high accuracy in the estimated relative state. Several IP algorithms have been proposed through the years to extract different types of features (corners and edges, mainly) and applied within the context of visual-based navigation, like in the case of the Hubble Space Telescope Servicing Mission 4 [42]. The interested reader can retrieve more details on non-AI-based feature detectors in [43–45]. Independently from the type of retrieved features, operating in spaceborne scenarios poses several challenges, such as varying illumination conditions, frequent self-occlusions, background celestial bodies, and varying size of the target in the camera Field of View (FOV). Despite the possibility of leveraging non-AI-based IP algorithms in spaceborne scenarios was assessed in [34], relying only on line segments, and in [46], leveraging a corner and two line segments detectors, the same works also demonstrate the high sensitivity of the involved IP algorithms to the previously mentioned challenges of spaceborne images.

The increasing interest in AI-based solutions of the last few years, particularly in deep learning and CNNs, also impacted the domain of

spacecraft relative pose estimation. Namely, the outcomes of SPEC2019 [26] demonstrated that relying on hybrid solutions involving CNN-based target detection (to reduce the effect of background bodies [21, 47]) and feature regression, coupled with classic PnP solvers, allows retrieving relative pose estimates with higher accuracy than both end-to-end AI-based methods [48,49] and non-AI-based algorithms. The main advantages offered by CNN-based IP algorithms include higher robustness against common disturbances like illumination condition variations, low signal-to-noise ratio, and presence of background bodies [21,50], as well as the possibility of retrieving only a predefined subset of all the detectable features, reducing the needed feature post-processing time with respect to non-AI-based solutions [51]. *UniAdelaide* [22] achieved the best performances on synthetic SPEED [23] images by leveraging a ROI extraction step based on Faster-RCNN [52] with HRNet as backbone [53] followed by a keypoint regression network entrusted to an HRNet [53] that retrieves the 2D location of 11 preselected landmarks from heatmaps of size 768×768 . The final relative pose is retrieved from the 2D-to-3D correspondences using a robust nonlinear optimization. Despite the high accuracy on synthetic images, this architecture struggles on mock-up images, where *EPFL_cvlab* achieved better performances. This architecture relies on a segmentation-driven CNN [54] to split the input image into fixed-size grids. Each grid predicts the presence of the target and determines the 2D locations of 8 keypoints alongside their confidence values. From the pool of keypoint candidates, the n most confident keypoints are selected to compute the pose estimate using a RANSAC-based PnP solver. Despite the improvements in the accuracy on mock-up images, also this architecture shows a huge performance gap between synthetic and mock-up images. A better balance of the performances on synthetic and mock-up frames was achieved by Piazza et al. [21] leveraging a ROI estimation entrusted to a YOLOv5s followed by a keypoint regression network based on HRNet-32 [53] that retrieves 11 heatmaps associated with preselected keypoints. The architecture in [21] also introduced an outlier rejection scheme based on the confidence associated with each keypoint that improved the accuracy on both synthetic and mock-up frames. Notably, the architecture in [33] registered performances on both synthetic and mock-up images close to those of Piazza et al. [21], addressing also the issue related to the deployability onboard by relying on lightweight CNNs. This architecture differentiates from the previous ones since, after the ROI estimation entrusted to the YOLOv5s, it leveraged line segments and keypoints extracted from a modified version of the M-LSD model [55], proving that high-level performances can also be achieved by relying on lightweight architectures, being among the lightest models that participated in SPEC2019 [29]. Despite the lightweight CNNs and the adoption of line segments as features to reduce the dimensionality of the search space as in [34], the main bottleneck of the M-LSD-based architecture is the need to process a high number of 2D-to-3D correspondences to retrieve the best relative pose, being the most time-consuming step, resulting in overall computational time that is compatible with the first pose initialization, but not with subsequent relative navigation filters update.

The domain gap bridging capabilities of pose estimation algorithms were further addressed in the SPEC2021 [27] with the SPEED+ dataset [24]. The objective of the competition was to evaluate the domain bridging capabilities by training the architectures on synthetic images and then testing them on two mock-up datasets named *Lightbox* and *Sunlamp*. *TangoUnchained* [27], developed by the same authors of *UniAdelaide*, achieved the best performances on the *Lightbox* dataset. This architecture resembles the *UniAdelaide* [22], improving the domain gap capabilities through a dedicated adversarial landmark regression model. Namely, the HRNet landmark regressor generates features fed to the ResNet34-based [56] discriminator module trained to identify if the landmarks come from a synthetic or a mock-up image, improving the accuracy of regressed keypoints [27] at the cost of an increased computational complexity of the overall pipeline of *UniAdelaide* that was already one of the most demanding [29], making the onboard

deployability issues more severe. Instead, concerning the Sunlamp dataset, the best performances have been registered by *lava1302* [27, 57]. Namely, the *lava1302* architecture consists of a single shared backbone based on HRFormer-S [58] followed by two heads made by ASPP blocks [59] that retrieve the heatmaps corresponding to pre-selected keypoints and perform part-wise segmentation, respectively, forcing the shared backbone to learn more robust features. Noticeably, also *lava1302* leverages a discriminator block trained via adversarial learning that takes as input the segmentation mask. An approach based on a shared backbone leveraging EfficientNet and BiFPN [60] was also adopted for the SPNV2 [31]. Differentiating from the previous architectures, the domain gap bridging capabilities of SPNV2 are mainly due to the multitask-learning and the synthetic augmentation adopted during the training phase to force the CNN to learn more robust features belonging to the target, independently from the context and domain, avoiding the complexities related to discriminator modules. Similarly, also the work in [61] adopted multitask-learning and synthetic augmentations to boost the performances of the Vision Transformer (ViT) architecture named PVSPE (Pyramid Vision multitask transformer network for Spacecraft Pose Estimation). Further, the ViT-based PVSPE features two novel feature extraction modules, i.e., the EnhancedPVT and the SlimGFPN, to combine global spatial and channel attention and to fuse features more effectively, respectively. Additionally, the PVSPE architecture relies on pose error losses computed using both the Matrix Fisher and the Multivariate Gaussian distributions that were revealed to be highly effective in reducing the overall relative pose error also on unseen images with a huge domain gap with respect to the training set, as proven by the scores achieved on the test sets of SPEC2021 [61].

Thus, the strategy adopted within this work to deal with computational and overall running time constraints, as well as improve the domain gap bridging capabilities, is to rely on lightweight, efficient, and multitasking CNNs to detect the target and regress preselected keypoints already ordered to provide a unique 2D-to-3D feature match, training those CNNs using synthetic augmentation strategies aimed at reducing the domain gap between synthetic training images and mock-up testing frames. Relying on CNNs to extract keypoints and estimate the relative pose of objects with respect to the camera is an active field of research, and, in the last years, most of the algorithms have been developed for the case of HPE. The HPE methods that rely only on single frames can be categorized depending on their capability of handling multiple targets [62]. If only a person is present within the image, the methods can be classified into regression-based and detection-based methods. Regression-based methods leverage an end-to-end approach to directly retrieve the joint coordinates of the body from the input image, while detection-based algorithms (as the HRNet [53] mentioned above) rely on independent heatmaps to detect each keypoint. Otherwise, if HPE shall be performed for multiple people in the images, it is possible to rely on top-down or bottom-up approaches [62]. The former methods first perform person detection and then retrieve the keypoints within each bounding box, while the latter ones work by predicting all the keypoints and then assigning them to different people. The interested reader is referred to [62] for a detailed overview of HPE methods, datasets, and evaluation metrics. Please notice that almost all the methods for keypoints regression that leverage a YOLO as target detectors (e.g., YOLO-pose [63]) can be classified as top-down approaches. Further, it is worth mentioning that top-down methods can also be adopted for a single target within the image, making it possible to have both the keypoints and the ROI as output.

Among the latest HPE methods released, there is the YOLOv8-pose architecture.² Similarly to the YOLOv5, the YOLOv8 model leverages a DarkNet53 backbone modified with intermediate C2f layers to reduce the computations while increasing the accuracy [64] (a similar

approach has been already introduced in YOLOv4 [35]). C2f layers are CSPNet (i.e., cross-stage partial connections) layers modified in the YOLOv8 implementation, combining high-level features with contextual information to improve detection accuracy. Further, a fast implementation of the spatial pyramid pooling [65] (named SPPF) has been included at the end of the backbone to accelerate computation by pooling features into a fixed-size map, enabling the model to capture both local details and broader context, increasing the robustness to variations in object size. Then, concerning the neck, an approach based on the feature pyramid network (FPN) [66] has been combined with a modified path aggregation network (PAN) [67], relying on the work already performed for the YOLOv4 [35]. The combination of these two approaches allows a bidirectional feature aggregation that enables an information flow from higher to lower levels and vice versa. Notably, concatenating the features bidirectionally through lateral connections in the top-down and bottom-up pathway creates fused feature representations that capture high-level semantic and detailed spatial information. This approach in the neck is beneficial in case the target varies in scale within images [67]. Lastly, detached heads are included to perform detection and pose estimation at different levels of the neck. The heads further process the inputs using convolutional layers that generate heatmaps processed to retrieve the final outputs, i.e., the bounding box and the class of the detected object, as well as the location of the keypoints and the associated confidence.

Due to the well-promising performances of YOLOv8-pose and its efficient architecture that strongly reduces the number of parameters with respect to most of the architectures that participated in SPEC2019 and SPEC2021 [29], enabling real-time inference on CPU [35], the YOLOv8-pose model has been adopted within this work to perform both ROI detection and keypoint regression for spaceborne applications.

3. Pose initialization algorithm

This section provides a detailed overview of the proposed algorithm for autonomous relative pose estimation. Namely, Section 3.1 details the developed algorithm, providing insights on the implemented steps to increase the robustness and reliability of the entire pipeline. Section 3.2 outlines the labeling procedure adopted for SPEED and SPEED+ to have annotations compliant with the adopted CNN model, providing details about the splits of these datasets and how they are involved in the training and testing phases. Lastly, Section 3.3 deals with the description of the YOLOv8-pose model and the introduced synthetic augmentations to bridge the domain gap between the synthetic training dataset and the actual mock-up frames representative of a real application scenario.

3.1. Architecture overview

The proposed YOLOv8-based relative pose initialization algorithm is schematized in Fig. 1.

The pipeline involves two tasks entrusted to the same CNN, i.e., the target detection and the keypoint regression. The full-scale image is processed by adding the noise (if not already included in the input images) and fed to the YOLOv8-pose. The CNN outputs the detected ROI, the regressed 2D keypoints, and the confidence level from 0 (low confidence) to 1 (high confidence) for each of them. Notably, the keypoints are returned always in the same order specified with the ground truth labels, making it possible to avoid searching for the admissible 2D-to-3D matches. A check on the necessity of a second inference through the same CNN, with the same weights and hyperparameters as the first inference, is performed once the outputs of the first YOLOv8-pose inference are available. Namely, the following checks are performed in sequence after the ROI diagonal size d_{ROI} and the aspect ratio $AR_{ROI} = \max(width_{ROI}, height_{ROI}) / \min(width_{ROI}, height_{ROI})$, as well as the keypoints confidence KP_{conf} , are retrieved:

² <https://docs.ultralytics.com>.

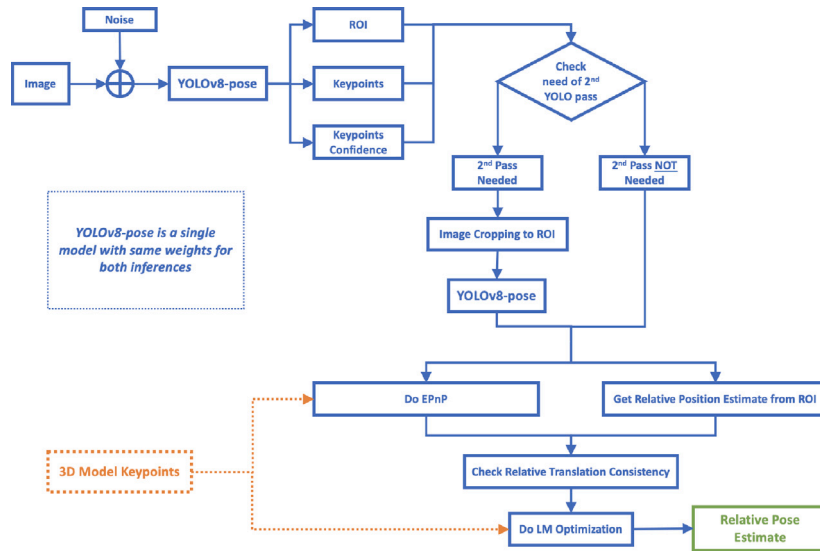


Fig. 1. YOLOv8-based pose initialization pipeline overview.

- if more than 30% of the keypoints have an associated confidence below a threshold $k_{kp, conf} = 0.45$, a second inference is needed to increase the accuracy of the retrieved keypoints;
- if one of the bounding box sides is overlapping (even partially) with one of the image borders and the $AR_{ROI} > 1.5$, the target is likely to be partially out of FOV, then a second inference is performed to increase the accuracy of the detected keypoints;
- if $d_{ROI} < k_{diag} \cdot d_{img}$, where k_{diag} is a multiplicative factor applied to the diagonal size of the image in pixels d_{img} , then a second inference is beneficial in refining the position of the keypoints detected due to the small size of the target in pixels compared to the full-scale image size.

The conditions and thresholds outlined above have been defined based on empirical observations and the hyperparameter optimization outcomes discussed in Section 4.3.1. Namely, concerning the first condition, preliminary tests showed that accurately localized keypoints are usually associated with high confidence values (above 0.8). This has been confirmed by the optimal confidence threshold for keypoints to be included in the EPnP solver, i.e., 0.88 (please refer to Section 4.3.1 for dedicated analyses). Consequently, a lower-bound threshold of 0.45 (approximately half the optimal confidence threshold) was adopted to identify unreliable keypoints. If more than 30% of the detected keypoints (i.e., a substantial amount of the overall keypoints) fall below this threshold, the pose reconstruction is likely to be affected by incorrect localization, and a second inference is performed to improve the keypoints accuracy. Similarly, the condition involving the ROI aspect ratio and border overlap is motivated by the observation that, for approximately cubic targets such as Tango, an aspect ratio greater than 1.5 combined with a partial overlap of the bounding box with the image borders often indicates that the object is partially outside the FOV. In this case, refining the inference on the visible region of interest enhances the confidence and spatial precision of the detected keypoints within the FOV, while reducing the confidence associated with keypoints related to portions of the target out of FOV. It is worth mentioning that this check on the ROI aspect ratio should be tuned or discarded if the actual target has a highly elongated shape, as it has been done in this paper for the case of Envisat discussed in Section 4.4. Lastly, the condition on the diagonal size of the ROI accounts for the relative scale of the target within the image. When the ROI is significantly smaller than the image (as defined by the factor k_{diag}), the target is assumed to be far from the camera and, in these cases, a second inference on the cropped region allows for more accurate

keypoint regression. The k_{diag} parameter is not defined a priori, being strongly related to the camera parameters and the target size, thus it is included in the hyperparameters optimization detailed in Section 4.3.1.

If one of the conditions above is verified, then the second inference is performed by feeding the image cropped to the ROI (retrieved during the first inference) to the same model of the first inference. Notably, the ROI and the keypoints coordinates retrieved from the second inference are translated to be expressed in the full-scale image reference frame before further processing.

The PnP problem is solved through the EPnP solver by using a subset of the available 2D keypoints and their corresponding 3D keypoints from the known 3D model of the target. It is pointed out that the EPnP is selected as the PnP solver due to its proven efficiency and robustness against noises in the estimated 2D keypoint location [34,39]. Further, it has been demonstrated in [68] that the EPnP features low processing time, especially for a low number of 2D-to-3D correspondences. Although these advantages with respect to other PnP solvers, the low robustness of the EPnP to outliers in the 2D-to-3D matches provided as input has been noticed in [39]. Consequently, outlier rejection schemes and optimal keypoint subset methods have been introduced through the pipeline to increase the overall robustness and avoid feeding the EPnP with bad correspondences. Namely, in agreement with [21], a minimal subset of keypoints is defined by picking a constant amount of keypoints N_{kp} in descending order of associated confidence score (i.e., the N_{kp} with the highest confidence score). Subsequently, all the detected keypoints with an associated confidence score higher than a threshold $\tau_{kp, conf}$ are appended to the minimal subset. Both N_{kp} and $\tau_{kp, conf}$ are included in the hyperparameters optimization outlined in Section 4.3.1. The EPnP gives as output the estimated relative pose, also used to compute the reprojection error E_{repr} defined as the mean Euclidean distance between the 2D keypoints and their corresponding 3D landmarks reprojected in the image using the estimated relative pose. Before proceeding with the optimization of the EPnP output through the Levenberg–Marquardt (LM) minimization scheme, the relative position from the EPnP t_{EPnP} is compared with a coarse estimate t_{coarse} retrieved by combining ROI and keypoints, to detect possible outliers, similarly to [21]. Namely, an outlier is detected if:

- the difference between the relative distances from the EPnP t_{EPnP} and from ROI and keypoints t_{coarse} is higher than 75%:

$$\frac{t_{EPnP} - t_{coarse}}{t_{coarse}} > 75\% \quad (1)$$

- the difference between t_{EPnP} and t_{coarse} is higher than 40% and E_{repr} normalized using the ROI diagonal size is higher than 7.5%:

$$\frac{t_{EPnP} - t_{coarse}}{t_{coarse}} > 40\% \quad \text{and} \quad \frac{E_{repr}}{d_{ROI}} > 7.5\% \quad (2)$$

- the difference between t_{EPnP} and t_{coarse} is higher than 15% and E_{repr} normalized using the ROI diagonal size is higher than 10%:

$$\frac{t_{EPnP} - t_{coarse}}{t_{coarse}} > 15\% \quad \text{and} \quad \frac{E_{repr}}{d_{ROI}} > 10\% \quad (3)$$

It is worth mentioning that the thresholds in Eqs. (1) and (3) were adopted from [21], while the condition in Eq. (2) was empirically defined to identify outliers falling within a moderate range of normalized distance mismatch errors but being more stringent on the geometric inconsistencies provided by the normalized reprojection error, effectively serving as an intermediate case between the extreme conditions addressed in Eqs. (1) and (3).

Whenever an outlier is detected, if the ROI sides are not partially overlapping with the image borders (meaning that the target can be partially out of FOV), the coarse relative position t_{coarse} is adopted as the initial relative position estimate fed to the LM optimization step coupled with the relative attitude given by the EPnP. Otherwise, the optimization step takes as input the relative pose given as output by the EPnP. The outcome of the LM optimization constitutes the final relative pose estimate retrieved from the proposed architecture. Notice that the LM optimization has been selected due to its proven superior performances with respect to other optimization schemes [21,33], e.g., the Newton–Raphson method used in [34].

3.2. Datasets and labels

The standard annotations for the YOLOv8-pose consist of a single text file for each image with labeled the ROI coordinates in the YOLO format and the coordinates in pixels of each keypoint (set to zero if out of the image) with a visibility score (0 if the keypoint is out of the FOV, 1 if it is inside the FOV but occluded, 2 otherwise) for each of them. The number of keypoints and their location have been selected in agreement with [21]. Namely, due to the simple geometry of Tango, only eleven keypoints are annotated, i.e., the four corners of the top and bottom panels (i.e., eight corners) and the tip of the three principal antennas. The visibility of each keypoint has been defined by exploiting the Möller–Trumbore ray-triangle intersection algorithm [69] leveraging the already labeled relative pose associated with each frame of the datasets, as outlined in [70]. The ROI can be computed for each image leveraging the 3D keypoints and the relative pose [33] for SPEED and SPEED+ images. The SPEED and SPEED+ datasets have been selected since they are the standard benchmark for relative pose estimation tasks.

The SPEED dataset splits are the same already adopted for the M-LSD-based pose initialization in [33], allowing a comparison between the two architectures. Namely, the splits comprise 7680 training images, 1920 validation images, and 2400 testing images. Since the images in the test set are the same already used in [33], the assessment of the representativeness of the labeled test set with respect to the SPEC2019 discussed in [33] has not been repeated. The SPEED+ synthetic images have been split into 47 968 synthetic images in the training set and 11 992 frames in the validation set. The test set for the SPEED+ dataset is constituted by the publicly available mock-up images and their associated relative pose [24], i.e., the *sunlamp* images (i.e., 2791 mock-up images acquired with HIL simulating the direct sun illumination) and the *lightbox* images (i.e., 6740 mock-up images acquired with HIL replicating the Earth albedo). Please notice that the frames in the test sets for both SPEED and SPEED+ datasets are never used during the training phase of the YOLOv8-pose model. It is worth

mentioning that the splits adopted for the SPEED+ dataset and the strategy of training only on synthetic images while testing on mock-up images are compliant with the SPEC2021. Hence, it is possible to consistently compare the outcomes of the proposed architecture with those that participated in SPEC2021 and to properly assess the domain gap bridging capabilities of the YOLOv8-pose model adopted here.

3.3. Target and keypoint detection CNN

The detector adopted within this work is the YOLOv8-pose due to its improved capabilities with respect to YOLOv5 in the target detection task and, remarkably, the possibility of regressing the keypoint location through a dedicated head inspired by [63]. The improved performances with respect to YOLOv5 are mainly due to the enhancement of the backbone model and the convolutions in the neck architecture that increased both the accuracy and efficiency of the inference. The YOLOv8-pose model was meant for HPE, and it has never been applied within the context of spacecraft relative pose estimation up to the author's knowledge. Despite that, human and spacecraft pose estimation entails almost the same foundation. Consequently, the only change made to the baseline implementation involved the relative weight associated with each keypoint that, for the case of spacecraft relative pose estimation, has been set equal for all of them and inversely proportional to the number of keypoints considered. Among the models of the YOLOv8-pose series, the YOLOv8s-pose (i.e., the *small* version) has been selected as the baseline since it offers a good trade-off between accuracy on benchmark HPE datasets and computational complexity. Namely, the YOLOv8s-pose model has 187 layers with a total of about 11.5M parameters and requires 29.5 GFLOPs during inference for an input image of size 512×512 pixels and 11 keypoints to be regressed. The input size has been selected equal to the YOLOv5s model adopted in [33] to allow a direct comparison of the performances.

3.3.1. Custom augmentations overview

By analyzing the gap in noise levels between mock-up images in the SPEED+ datasets and SPEED synthetic frames, it has been conjectured that a more accurate VIS sensor model applied to synthetic images used during the training phase can enhance the performances of the CNN in testing on mock-up frames. Consequently, a dedicated new augmentation named *VIS sensor augmentation* has been developed leveraging the high-fidelity VIS sensor model introduced in [51]. Namely, this new augmentation introduced here relies on a more representative noise applied to each image following the scheme reported in Fig. 2. The developed VIS sensor noise model merges the approaches in [71,72] and the Fixed Pattern Noises models retrieved from the EMVA1288 standard [73]. Namely, the input photons received by the sensor are estimated by applying the ADC Gain and the Quantum Efficiency to the image processed. The estimated photons are then processed by including the noise sources and scaling factors outlined in Fig. 2 (whose hyperparameters are defined starting from the datasheet of the assumed camera), giving as output the noised image. More details on the modeling assumptions of each noise source and a comparison with frames from an actual CMOS sensor are available in [51]. The model in Fig. 2 has been adopted as synthetic augmentation by randomizing all its hyperparameters (i.e., the assumed camera characteristics reported in EMVA1288-compliant datasheets) for each processed image to avoid overfitting the CNN on a single noise level.

Further, the possibility of augmenting the training frames by adding synthetic flares and blooming has been evaluated to train on images that resemble the SPEED+ mock-up images [31]. The generation of synthetic flares leverages a dedicated tool available in the *Albumentation* package [74] that has been modified within this work to apply the flares only on the brightest region of the target (eventually serving as a pseudo-erase augmentation since, depending on the radius of the flare, portions of the images became completely saturated) and, additionally, smoothing them to avoid the ringing phenomena that

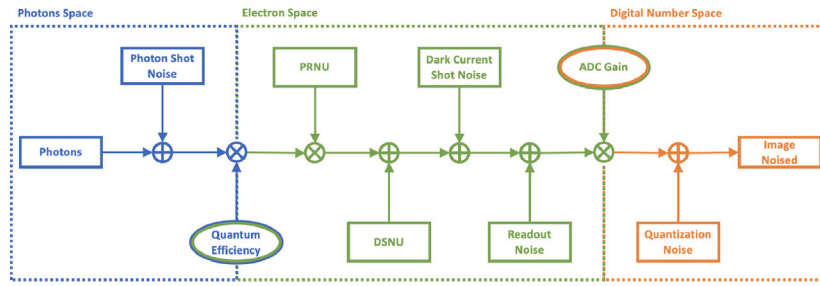


Fig. 2. Scheme of synthetic VIS sensor noise model.

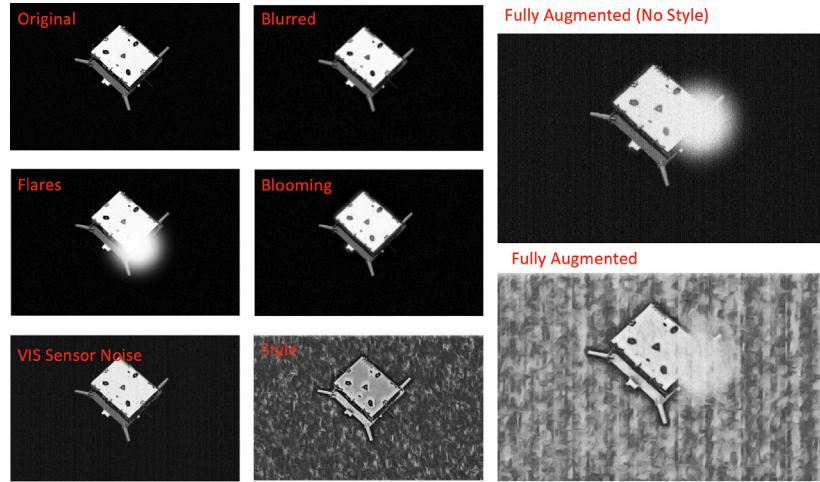


Fig. 3. Visualization of the augmentations adopted in the YOLOv8s-pose training.

are not representative of actual flares. Blooming is intended here as the haze effect noticed on mock-up images in correspondence with bright regions. The blooming is applied using a custom routine outlined for the work presented here that defines for each image processed a binary mask containing the pixels having an intensity higher than a threshold. The mask isolates the bright pixels in the original image. Both the mask and the brightest pixels are blurred in an iterative way. The bloomed image is defined by blending the original image and the blurred brightest pixels using the blurred mask as weight. Please notice that the threshold values for the mask definition and the number of blurring iterations are randomized in the augmentation developed. The last augmentation evaluated is a style augmentation [75] (or texture randomization) that randomizes the textures within the images via neural style transfer [76], limiting the focusing of the CNN on the local appearance of synthetic textures that may not replicate the actual aspect of the mock-up model. The style augmentation was included due to its proven enhancement in the domain gap bridging capabilities of the trained model [31], despite the high amount of computational resources required by this last augmentation. A visualization of the augmentations discussed above is given in Fig. 3.

The YOLOv8s-pose model has been trained on synthetic SPEED images, adding the augmentation strategies described in this section to the baseline YOLOv8 augmentations to assess the capabilities of bridging the domain gap between synthetic and mock-up images, as discussed in Section 4.2.2 dedicated to the training and ablation study of the YOLOv8s-pose model.

4. Results

4.1. Evaluation metrics

The evaluation metrics adopted to assess the performances of the YOLOv8s-pose model after the training on the synthetic SPEED images and the pose estimation performance metrics are detailed hereafter.

CNN performance metrics. The criteria adopted to evaluate the performances of YOLOv8s-pose are the Intersection-over-Union (IoU) and the Average Precision (AP) for the object detection performances, while the Object Keypoint Similarity index (OKS) and the keypoint AP are considered as the keypoint detection performance metrics. The definition of these standard metrics is not reported here for brevity, but the interested reader can refer to [62] for a detailed overview of them. Notably, the OKS for keypoints is almost equivalent to IoU for bounding boxes. Hence, it is possible to define the Average Precision (AP) for the keypoints as the area under the precision–recall curves generated by adopting different threshold levels for the OKS. Even for that case, it is possible to define the AP_{50}^{95} as the mean of the AP values computed for precision–recall curves obtained by thresholding the OKS to 50% and 95%. Similarly to the case of object detection, the higher the AP_{50}^{95} , the more accurate the keypoint detection.

Pose estimation performance metrics. The metrics adopted to estimate the accuracy are those defined for SPEC2019 and SPEC2021 [26,27], i.e., the relative position error E_t (Eq. (4)) and its normalized value e_t (Eq. (5)), the relative quaternion error e_q (Eq. (6)), and the SLAB score e_{SLAB} (Eq. (7)) that can be defined for N frames.

$$E_t = \|E_t\| = \|\mathbf{t}_C - \hat{\mathbf{t}}_C\| \quad (4)$$

$$e_t = \frac{E_t}{\|\mathbf{t}_C\|} = \frac{\|\mathbf{t}_C - \hat{\mathbf{t}}_C\|}{\|\mathbf{t}_C\|} \quad (5)$$

$$e_q = 2 \cdot \arccos |\mathbf{q} \cdot \hat{\mathbf{q}}| \quad (6)$$

$$e_{SLAB} = \frac{1}{N} \sum_{i=1}^N \left(e_t^{(i)} + e_q^{(i)} \right) = \frac{1}{N} \sum_{i=1}^N \left(\frac{\|\mathbf{t}_C^{(i)} - \hat{\mathbf{t}}_C^{(i)}\|}{\|\mathbf{t}_C^{(i)}\|} + 2 \cdot \arccos |\mathbf{q}^{(i)} \cdot \hat{\mathbf{q}}^{(i)}| \right) \quad (7)$$

where $\mathbf{t}_C$ and $\mathbf{q}$ are the ground truth pose vectors, while $\hat{\mathbf{t}}_C$ and $\hat{\mathbf{q}}$ are the estimated ones. Further, the per-axis angular error $\mathbf{E}_\theta = [\theta_x, \theta_y, \theta_z]$ [33], representing the angular estimation error of each axis of the target body reference frame with respect to the camera reference frame, is also included and defined as in Eq. (8).

$$\theta_i = \arccos(\mathbf{r}_{est,i} \cdot \mathbf{r}_{GT,i}) \quad \text{with } i = 1, 2, 3 \quad (8)$$

where $\mathbf{r}_{est,i}$ and $\mathbf{r}_{GT,i}$ are the i th columns of the estimated relative attitude $\mathbf{R}_{est} = \mathbf{R}_{TRG \rightarrow CAM}^{est}$ and the ground truth $\mathbf{R}_{GT} = \mathbf{R}_{TRG \rightarrow CAM}^{GT}$ respectively.

4.2. YOLOv8s-pose training and ablation study

The YOLOv8s-pose training parameters and the ablation study performed to assess the effects of the introduced augmentations and select the best model are discussed in Sections 4.2.1 and 4.2.2, respectively.

4.2.1. Training parameters

The baseline model (i.e., the YOLOv8s-pose) is trained from scratch on the synthetic SPEED images using the default augmentations of the YOLOv8 training such as random brightness variations, CLAHE (Contrast Limited Adaptive Histogram Equalization), blurring, and mosaic. Notably, the mosaic augmentation is deactivated for the last ten epochs to fine-tune the architecture to the case of a single target within the scene. The default blurring of the YOLOv8 training has been modified here to select, with the same probability, a blurring algorithm among Gaussian blur, median blur, and motion blur. The training is performed on an NVIDIA RTX™ A6000 GPU using the stochastic gradient descent method with a learning rate of 0.01 linearly decaying to 0.0001, momentum of 0.937, weight decay of 5×10^{-4} , and a minibatch size of 64 images. The overall loss function is the sum of the object and keypoint detection losses. The object detection loss entails a binary cross entropy for the classification problem and a bounding box loss that leverages the Complete Intersection-over-Union (IoU) metric (i.e., it measures the overlap between predicted and ground truth bounding boxes but also considers the difference in aspect ratio, center distance, and box size) to measure the difference between the predicted and the ground truth bounding boxes. The keypoint detection loss encompasses the keypoints loss (based on the distance between the predicted and ground truth landmarks) and the keypoints object loss, i.e., a binary classification loss that classifies if a keypoint is present. The object detection and the keypoint detection losses are multiplied by their relative gains, set within this work to 7.5 and 18, respectively, and summed to define the overall loss function. The gains have been defined during a testing phase, being an optimal trade-off between object detection and keypoint regression performances on SPEED images. The YOLOv8s-pose training lasts for 900 epochs. The parameters discussed above are selected since they offer the best performances on SPEED images, while the minibatch size maximizes the number of frames in each mini-batch without surpassing the computational resources available. The number of epochs has been defined by noticing no meaningful improvements from training longer.

4.2.2. Ablation study and model selection

An ablation study to select the best-performing model has been performed by training from scratch the YOLOv8s-pose on synthetic SPEED images and gradually adding the augmentations outlined in Section 3.3.1. The VIS sensor noise augmentation has been applied to all the images. Instead, the random flare and blooming augmentation and the style randomization are triggered with a probability of 75% and 50%, respectively. The trained models have been evaluated in terms of mean IoU, mean OKS, and their respective AP_{50}^{95} on the test images from SPEED, SPEED+ Lightbox, and SPEED+ Sunlamp. The outcomes of this ablation study are reported in Tables 1 and 2, where the model M1 (i.e., the original baseline) is trained from scratch with the default augmentations and the hyperparameters already discussed

in Section 4.2.1. The outcomes of the ablation study point out that the VIS sensor noise augmentation effectively increases all the evaluated metrics. Namely, the model M2 trained by including the VIS sensor noise augmentation is the best-performing model on synthetic images. Also, the VIS sensor noise augmentation brings the most intense improvements to the object detection metrics for the mock-up images. Concerning the metrics about the keypoint regression, the VIS sensor noise is highly effective both in synthetic and mock-up images from the SPEED+ Lightbox dataset, but the addition of the *Blooming and Flare* augmentation in model M3 gives the best improvement on the SPEED+ Sunlamp dataset. This outcome is expected since visible intense flares (due to the simulated direct Sun illumination) characterize the Sunlamp frames. Notably, the best-performing model on mock-up images is model M4, i.e., trained with all the augmentations, including the style augmentations. While the effects of the style augmentation are not strong on the object detection and in the keypoint regression for synthetic and Lightbox images, this augmentation revealed particularly helpful in increasing the keypoint regression statistics in Sunlamp images. The slightly worse performances on synthetic images highlighted for models M3 and M4 are due to the domain gap between the augmented frames and those in the synthetic test dataset that do not show any flares or texture changes with respect to the non-augmented training images. Notably, the top performances achieved by model M2 on synthetic images point out that training with a randomized noise higher and more complex than the actual noise already included in the images is beneficial in increasing the overall accuracy. Consequently, the strategy adopted for model M4 shall be preferred whenever a model is expected to deal with images acquired with HIL (also in the absence of flares). Fig. 4(a) gives a visualization of the ROI predicted by model M4 on both synthetic and mock-up images, while Fig. 4(b) shows the regressed keypoints plotted with their confidence mapped in the color bar. Notice that landmarks not within the FOV have extremely low confidence scores associated and, consequently, the outlier rejection strategy outlined in Section 3.1 can easily reject them from the subset adopted to solve the PnP problem.

As a side result, Table 3 provides a comparison of the target detection metrics for SPEED images scored by the architecture proposed here and other methods available in the literature, proving that the YOLOv8s-pose model M2 is the best performing in target detection tasks, with highest IoU score on SPEED images (+1.1% than the YOLOv5s adopted in [33], and +2.2% than the SLN from [21]) and mean AP_{50}^{95} (+3.3% than the YOLOv5s adopted in [33], +0.5% than the SLN from [21]). Notably, even the models M3 and M4 perform better than the CNNs listed in Table 3. Moreover, by comparing the keypoint regression mean AP_{50}^{95} scored by the HRNet32 adopted in [21] ($AP_{50}^{95} = 98.97\%$), i.e., the best keypoint regressor on SPEED images available in the literature, with the ones achieved by models M2, M3, and M4 ($AP_{50}^{95} = 99.40\%$), it arises that the YOLOv8s-pose is also the best-performing model on keypoint regression task for SPEED images, up to the author's knowledge. These side results highlight that the multi-task learning and the adopted augmentations effectively enhance the model performances, outperforming CNNs trained and applied only on a single specific task on synthetic images.

4.3. Results on benchmark datasets

The results on benchmark datasets, i.e., SPEED labeled test set (Section 4.3.1) and both SPEED+ Lightbox and SPEED+ Sunlamp (Section 4.3.2) are detailed hereafter.

4.3.1. SPEED

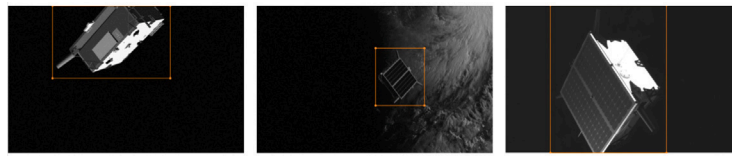
The model M4 has been adopted to evaluate the scores of the YOLOv8s-pose-based pose estimation algorithm on the labeled test set extracted from SPEED since it is the model that best generalizes even to mock-up images, despite not being the best performing one on synthetic images.

Table 1
Ablation study for YOLOv8s-pose augmentations, object detection results.

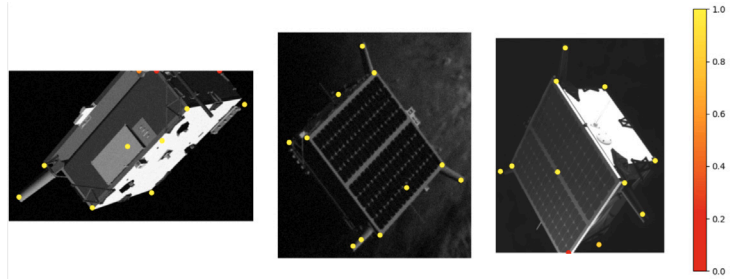
Model	Augmentation	SPEED		SPEED+ Lightbox		SPEED+ Sunlamp	
		ROI mAP_{50}^{95}	IoU mean	ROI mAP_{50}^{95}	IoU mean	ROI mAP_{50}^{95}	IoU mean
M1	Baseline	0.981	0.9653	0.687	0.8280	0.627	0.8349
M2	+ VIS sensor noise	0.990	0.9758	0.843	0.8939	0.723	0.8682
M3	+ Blooming and flare	0.989	0.9743	0.874	0.9041	0.802	0.8826
M4	+ Style augmentation	0.988	0.9723	0.881	0.9064	0.831	0.8900

Table 2
Ablation study for YOLOv8s-pose augmentations, keypoint regression results.

Model	Augmentation	SPEED		SPEED+ Lightbox		SPEED+ Sunlamp	
		Keypoints mAP_{50}^{95}	OKS mean	Keypoints mAP_{50}^{95}	OKS mean	Keypoints mAP_{50}^{95}	OKS mean
M1	Baseline	0.987	0.9747	0.472	0.6267	0.185	0.3928
M2	+ VIS sensor noise	0.994	0.9895	0.634	0.7226	0.262	0.4587
M3	+ Blooming and flare	0.994	0.9887	0.731	0.7728	0.422	0.5652
M4	+ Style augmentation	0.994	0.9873	0.770	0.8032	0.554	0.6558



(a) ROI detected on synthetic and mock-up images.



(b) Keypoints regressed in synthetic and mock-up images with confidence scores mapped in the colorbar.

Fig. 4. Examples of ROI and keypoints detected by model M4 on synthetic and mock-up images. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 3
Performance comparison of target detection CNNs on SPEED images.

Method	Mean IoU	Mean AP_{50}^{95}
SLAB baseline [77]	91.9%	N.A.
UniAdelaide [22]	95.34%	N.A.
SLN (YOLOv5s) [21]	95.38%	98.51%
YOLOv5n [33]	95.42%	95.7%
YOLOv5s [33]	96.46%	97.6%
YOLOv8s-pose M1 (our)	96.53%	98.1%
YOLOv8s-pose M2 (our)	97.58%	99.0%
YOLOv8s-pose M3 (our)	97.43%	98.9%
YOLOv8s-pose M4 (our)	97.23%	98.8%

Hyperparameters selection. The hyperparameters of the architecture comprise the scale factor k_{diag} applied to the image diagonal size when checking the need for a second inference, the minimum number of keypoints N_{kp} to form the minimum set of correspondences, and the confidence threshold $\tau_{kp,conf}$ to define which keypoint to add to the minimum subset before solving the EPnP problem. These hyperparameters are tuned to achieve the best performances in terms of mean SLAB score on the test dataset, similarly to [21], leveraging a 3D grid-search optimization. The value for k_{diag} has been evaluated within the range from 0.1 (one inference for all the images) to 0.7 (two inferences

for all the frames). N_{kp} ranges from 4 to 10 since the EPnP solver needs at least four 2D-to-3D matched keypoints to retrieve a unique solution. Notice that the upper limit is not 11 landmarks (i.e., all the keypoints adopted in this implementation) since this case is covered by tuning the confidence threshold $\tau_{kp,conf}$. The confidence threshold should be evaluated on the range from 0 to 1, i.e., its entire range of variability, but the experiments revealed that the results are not affected by thresholds below 0.6. Hence, the results shown are only for $\tau_{kp,conf}$ in the range between 0.6 and 1.0. Fig. 5 shows the outcomes of the 3D grid search performed to identify the best combination of hyperparameters. Namely, Fig. 5(a) shows the 3D grid defined during the optimization, while Fig. 5(b) provides the detail on how N_{kp} and $\tau_{kp,conf}$ affect the mean SLAB error for a fixed k_{diag} set to its optimal value. The retrieved optimal values are $k_{diag} = 0.3$, $N_{kp} = 4$, and $\tau_{kp,conf} = 0.88$. Fig. 5(a) points out that using only a single inference ($k_{diag} = 0.1$) is not effective, leading to a mean SLAB score of about 0.042, but also relying always on two inferences is a sub-optimal solution for the overall accuracy. The detailed plot for fixed $k_{diag} = 0.3$ shown in Fig. 5(b) highlights that always relying on all the keypoints regressed (i.e., $N_{kp} = 10$) is not optimal since potential outliers may affect the pose estimation process (e.g., for the case of target partially outside of the FOV). The optimization revealed that the optimal choice for the SPEED dataset is to set N_{kp} as the minimum number needed

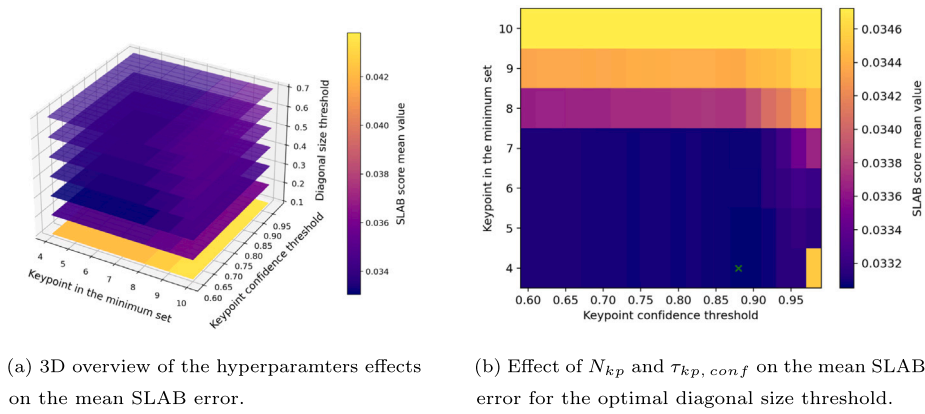


Fig. 5. Grid-search optimization of the hyperparameters in the architecture.

to retrieve a solution from the EPnP, ensuring that a relative pose is always estimated, and then add the keypoints with high confidence to this subset to increase the accuracy of the retrieved solution, discarding possible outliers identified by a low confidence score. Despite that, it is worth mentioning that the mean SLAB score in Fig. 5(b) is always better than the score achieved in [21,33] and its variability is extremely low. Consequently, a random selection of N_{kp} and $\tau_{kp,conf}$ will not jeopardize the overall architecture performances.

Pose initialization performances. The error metrics evaluated for the implemented architecture on SPEED synthetic images are reported in Table 4, while Fig. 6 shows the histograms of the error distributions in the 3σ range. The proposed architecture achieved a mean SLAB score of 0.03305 with a standard deviation of 0.01979. This outcome denotes that the architecture is highly accurate and precise, resulting within the top-3 architectures on the official SPEC2019 leaderboard and top-10 architectures in the postmortem competition.³ The relative translation error is in the order of a few centimeters with a low standard deviation. Namely, the highest contribution to the overall error is given by the relative position error along the camera boresight axis (i.e., the z-axis) being about one order of magnitude higher than the error components along the x and y axes. This is a well-known behavior [21,33] and, for YOLOv8s-pose architecture, it arises from a slightly inaccurate estimation of the coordinates of the landmarks, resulting in a sort of “zoom-in/out” effect. Namely, if the keypoints coordinates are all retrieved with a displacement towards the outside direction from the target center of mass, the retrieved relative position will have almost zero error for the x and y components, while the z component will be underestimated (i.e., zoom-in effect). Additionally, whenever the relative position estimated from ROI and keypoints is applied to correct outliers in the output of the EPnP, the solution has a higher degree of uncertainty along the z-axis since it is computed by approximating the actual upper panel diagonal size of the target with an estimate that may not be completely accurate in representing the dimensions of the upper panel in pixels. Notice that all the keypoints detected shall be displaced along the same direction to give a meaningful error contribution along the x and y axes. Concerning the relative attitude error, the implemented architecture achieves a mean accuracy of about 1.40 degrees with an extremely low standard deviation of 1 degree, highlighting the high accuracy and precision of the proposed pipeline in the relative attitude estimation task. The values of E_θ show that the angular error is well distributed along the three axes with comparable means and standard deviations, despite the z-axis component being slightly higher than the others (also by looking at the histogram in Fig. 6). It is conjectured that this behavior is due to the selected 3D keypoints and because the final pose retrieved is the one that minimizes

the reprojection error, as already noticed in [21]. Namely, the set of selected 3D keypoints has a higher inertia associated with the z-axis [21], and the reprojection error for keypoints far from the geometric center of the target is higher than the error for a keypoint close to the center for the same angular error. Hence, the optimization scheme is biased to return relative pose estimates that optimally fit landmarks further from the center due to their higher contribution to the reprojection error, leading to a higher uncertainty of the relative attitude about the direction associated with the keypoints highest inertia, i.e., the target z-axis [21].

Effect of number of inferences and relative distance. The SLAB score for each image as a function of the ground truth relative distance is shown in Fig. 7, where the number of inferences performed is mapped on the color associated with each point. The plot shows that a single inference is performed up to a relative distance of about 8 m. Notably, the images generated within the relative distance range [0, 8] m are not always processed with a single inference due to the effects of the checks on the target partially out of FOV and on the keypoints confidence. Further, the plot excludes the presence of error peaks and strong outliers in the SLAB score for all the images, except for the two cases at about 4 m and 25 m, where the SLAB score reaches the maximum value of 0.43. Additionally, the absence of high SLAB errors for images at low distances confirms that the proposed architecture can handle the case of a target partially out of FOV, returning accurate relative pose estimates. The trend of the relative errors as a function of the ground truth relative distance is better detailed in Fig. 8, where the relative translation and attitude error are shown as a function of the increasing ground truth relative distance. The plots are generated by sorting the inputs based on the ground truth relative distance, grouping the outcomes in 30 batches of 80 images each, and then plotting the mean error of each batch in correspondence with the mean ground truth distance in the related batch. The blue-shaded region is the interquartile range of error corresponding to a 1σ range that represents the central trend of the error distribution, while the gray-shaded area indicates the full error range. The plots do not show the spiking behavior noticed in [21,33], confirming the absence of severe outliers that drive only the mean value, only slightly affecting the central trend. Further, the lack of spikes in the relative attitude error plot denotes the absence of wrong estimates due to geometric ambiguities. Both the relative position and attitude errors increase as the relative distance increases. The mean relative distance error has a higher increment rate as a function of the ground truth distance than the relative attitude, more pronounced for distances higher than ~ 18 m. The same holds for the uncertainties given by both the shaded areas. Despite that, it is worth pointing out that the highest mean error (i.e., ~ 30 cm) obtained for a ground truth distance of about 27 m corresponds to about the 1.1% of error. On the contrary, the relative attitude is only slightly affected since the mean value mildly increases for ground truth distances higher than ~ 20 m, but the central

³ Updated at 17/03/2024.

Table 4
YOLOv8s-pose error metrics scores on synthetic SPEED images.

Metric	Mean value	Standard deviation
E_t [cm]	9.51	11.14
E_r [cm]	[0.12, 0.11, 1.69]	[0.72, 0.74, 14.52]
e_t [–]	$[0.143, 0.151, 1.889] \times 10^{-3}$	$[0.083, 0.088, 1.078] \times 10^{-2}$
e_θ [deg]	1.40	1.00
E_θ [deg]	[1.08, 1.08, 1.12]	[0.89, 0.79, 0.95]
e_{SLAB} [–]	0.03305	0.01979

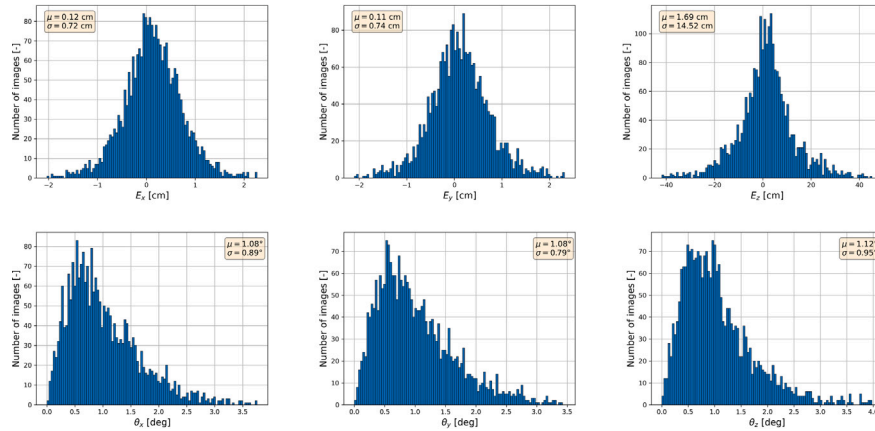


Fig. 6. Errors distributions in 3σ range for synthetic SPEED images.

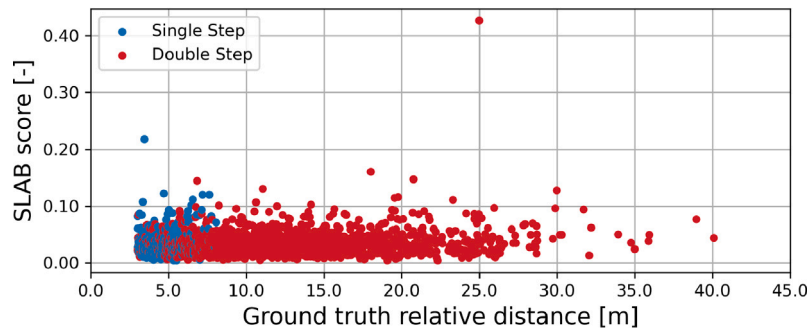


Fig. 7. Effects of the number of inferences on the SLAB score. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

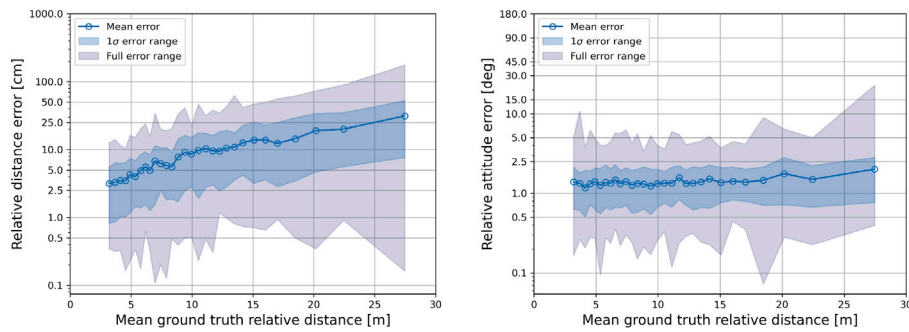


Fig. 8. Effects of relative distance on E_t (left) and e_θ (right) for SPEED synthetic images, y-axis in logarithmic scale. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

trend of the error (i.e., the width of the blue shaded area) is almost constant for the entire dataset.

Effect of image background. The effects of the Earth that can be present in the background of the synthetic SPEED images are evaluated by plotting the SLAB score in ascending order and differentiating the color depending on the background of the image as in Fig. 9. The presence

of the Earth has an almost null effect since the images with non-black backgrounds are distributed uniformly for about 98% of the dataset, with SLAB scores comparable with those achieved from images with a black background. The remaining 2% of the frames, i.e., those with the highest SLAB scores and the steepest increment of the error, mainly contain images with the Earth in the background. The worst estimate

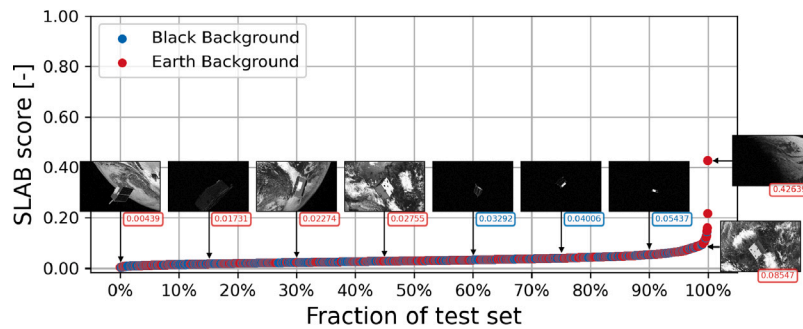


Fig. 9. Effects of image background on the SLAB score for synthetic SPEED images. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

($e_{SLAB} \approx 0.43$) is due to a combination of high relative distance and a target close to the terminator line and in low illumination conditions. At least one of these disturbances can be retrieved for the remaining frames in the last 2% of the dataset that exhibit the steepest increment of the error. Notice that the main effect of the disturbances mentioned above is on the reduced accuracy of the 2D location of the regressed keypoints, while the ROI is always retrieved correctly without cropping out portions of the target.

Runtime evaluation. The running time of the YOLOv8s-pose-based architecture has been evaluated for a Python-based implementation running on an Apple® Silicon™ M1 Pro, using the CPU only, measuring it for all the images in the test set. Fig. 10 provides the histogram of the total computational time registered through the entire dataset. The histogram plot points out two peaks in the computational time distribution due to the cases in which the pipeline performs just a single inference (corresponding to the peak at about 60 ms) and when the second pass through the YOLOv8s-pose is needed (corresponding to the second peak located at about 131 ms). Namely, the mean inference time is about 59 ms for the first inference and about 72 ms for the second one, while the average running time to retrieve the pose after the YOLOv8s-pose inferences (including the solution of the EPnP) is about 0.77 ms. The computational time for the case of two inferences has a higher variability due to the non-maximal suppression step needed to retrieve the final outputs of the model. Namely, in the cropped image, the CNN may extract more bounding box and keypoint location coarse estimates than in the full-scale image since the target occupies almost entirely the FOV and, additionally, the noise is more relevant due to the low amount of pixels. This high amount of coarse estimates is processed with a non-maximal suppression step to retrieve the final outputs, leading to a higher variability of the computational time. Notably, this architecture achieves computational times on CPU that are by far lower than the M-LSD-based architecture [33], which

leveraged extremely lightweight CNNs coupled with a more complex features post-processing algorithm that led to a minimum mean runtime of ~ 1.42 s on the CPU of an Apple® Silicon™ M1 Pro, proving that entrusting the 2D-to-3D matching to the CNN adopted here and making it possible to run the EPnP only once per image is effective in reducing the computational time. Notably, the YOLOv8s-pose inference time is lower than the mean values of ~ 85 ms for the YOLOv5s and ~ 88 ms for the MLSD reported in [33] due to its more efficient structure despite having slightly more parameters and requiring more GFLOPs. Additionally, the reduced time needed to recover the pose using only one inference can be exploited in an operational scenario to increase the update frequency of the navigation filters at low relative distances (i.e., when most of the images require a single inference), increasing the robustness and accuracy of the overall navigation pipeline.

For completeness, it is worth highlighting that a detailed runtime comparison with other methods could not be performed due to the absence of publicly available architectures for the models referenced in the literature. Moreover, a direct comparison based on existing studies is infeasible because the runtime evaluations were conducted on different hardware configurations, making them incompatible with the setup used in this work. This limitation has already been noted in [29] as a common shortcoming of the currently available solutions. To address this issue, [29] proposed using the number of model parameters as a first approximation for indirectly evaluating latency, memory usage, computational cost, and power consumption in relative pose estimation pipelines. This approximation is valid under the assumption that the inference step of the CNNs is the most time-intensive part of the pipeline (e.g., this does not hold for architectures like YOLOv5s/MLSD, where feature processing via EPnP dominates runtime [33]). Specifically, models with fewer parameters are expected to exhibit lower runtime. Therefore, in the subsequent sections, only the number of parameters for the compared models, where available, will be reported to provide insights into their expected runtime.

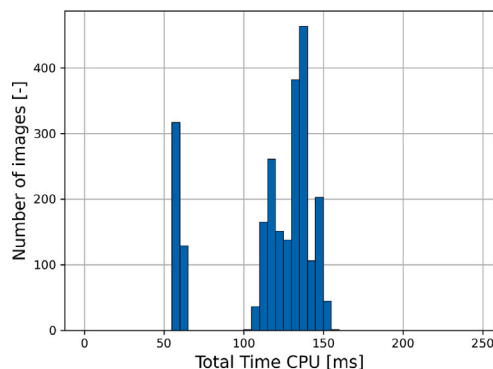


Fig. 10. Runtime histogram of YOLOv8s-pose-based architecture on synthetic SPEED images.

Comparison with other methods. A comparison of the performances and the number of parameters of the proposed architecture against the best models that participated in SPEC2019 is reported in Table 5. The architectures are compared using the mean value and standard deviation for the translation error E_t , the quaternion error e_q , and the SLAB score on synthetic images from SPEED. Please notice that the overall standard deviations are not available for the RPEP architecture from [21] hence, the highest standard deviation among the available values per axis is reported to give an approximate estimation of the overall value. Further, the number of parameters have been retrieved from [29,33,49]. The comparison points out that only *UniAdelaide* and *EPFL_cvlab* achieve better SLAB error scores than the YOLOv8s-pose-based pipeline, which outperforms all the other models. The translation and rotation errors highlight that the proposed architecture is the most precise in retrieving the relative angular position, having the lowest standard deviation. Regarding the relative translation, the standard deviations achieved by the YOLOv8s-pose ranked second, after the

Table 5

Performance comparison of YOLOv8s-pose architecture against other methods on SPEED images.

Architecture	Parameters (M)	E_t , cm ($\mu \pm \sigma$)	e_q , deg ($\mu \pm \sigma$)	e_{SLAB} (synthetic images)
UniAdelaide [22]	~49.8	3.2 ± 9.5	0.41 ± 1.50	0.0094
EPFL_cvlab [54]	~59.1	7.3 ± 58.7	0.91 ± 1.29	0.0215
massimo.piazza (RPEP) [21]	~36.1	10.36 ± 30.44	2.24 ± 10.82	0.0450
YOLOv5s/M-LSD [33]	8.5	8.61 ± 39.8	2.19 ± 12.59	0.0455
pedro_fairspace [49]	~500	14.5 ± 23.9	2.49 ± 3.02	0.0571
SLAB Baseline [47]	11.17	20.9 ± 113.3	2.62 ± 2.09	0.0626
YOLOv8s-pose M4 (ours)	11.5	9.51 ± 11.1	1.4 ± 1.00	0.0330

Table 6

Object and keypoint detection metrics for YOLOv8s-pose model fine-tuned on synthetic SPEED+ images.

Model	SPEED+ Lightbox				SPEED+ Sunlamp			
	ROI mAP_{50}^{95}	Keypoints mAP_{50}^{95}	IoU mean	OKS mean	ROI mAP_{50}^{95}	Keypoints mAP_{50}^{95}	IoU mean	OKS mean
M4	0.8821	0.770	0.9064	0.8032	0.831	0.554	0.8900	0.6558
M4 fine-tuned	0.936	0.88	0.9354	0.8873	0.879	0.734	0.9077	0.7712

architecture by *UniAdelaide*. Notably, the mean translation error of the M-LSD-based algorithm is lower than the YOLOv8s-pose value. Despite that, the standard deviations for the relative translation and rotation errors of the YOLOv8s-pose-based are respectively 4 times and more than 10 times lower than the corresponding scores registered for the M-LSD-based architecture, proving the higher reliability of the pipeline proposed here. Notably, both *UniAdelaide* and *EPFL_cvlab* adopted architectures with ~ 50M and ~59M parameters, respectively, [29], justifying the differences of performances if compared to the YOLOv8s-pose-based architecture, that is by far the smallest among the three. It is acknowledged that the *YOLOv5s/MLSD* and *SLAB baseline* architectures feature a lower number of parameters than the solution proposed here. Despite that, it shall be noticed that the SLAB baseline architecture achieved worse performances than the model proposed here, while the *YOLOv5s/MLSD* architecture has only slightly worse performances but by far higher runtime than the architecture proposed here, as already discussed in the previous paragraph. It is worth mentioning that all the architectures in Table 5, including the YOLOv8s-pose-based models, achieve 100% of solution availability.

4.3.2. SPEED+

The results achieved by the proposed YOLOv8s-pose-based architecture on mock-up images from SPEED+ are discussed here. The model trained on the synthetic SPEED images using all the augmentations (i.e., model M4 in Tables 1–2) has been fine-tuned for the SPEED+ images (using only the synthetic images of the SPEED+ dataset) via transfer learning, performing a new training of 100 epochs starting from the weights of model M4 and without changing the training hyperparameters. This step was performed to compensate for the high difference in the relative distance distribution between the images in SPEED and those in SPEED+. Table 6 reports the metrics scored by the fine-tuned model on the SPEED+ mock-up images and, to ease the comparison, the metrics previously reported in Tables 1–2 for the model M4. By comparing the scores achieved by the fine-tuned model M4 with those achieved by model M4 (trained only on SPEED images), it can be concluded that the fine-tuning is highly effective in increasing all the metrics. Hence, this comparison points out that it is of paramount importance to train the model on a dataset that at least resembles the relative distance distribution of the application scenario to achieve the best performances.

SPEED+ lightbox images. Examples of images from the SPEED+ Lightbox dataset are shown in Fig. 11. The relative pose estimation performances of the YOLOv8s-pose architecture for this dataset are summarized in Table 7. As expected, the YOLOv8s-pose-based architecture performances are worse on mock-up images due to the domain gap between the synthetic training images and the mock-up testing frames.

Table 7

YOLOv8s-pose error metrics scores on SPEED+ Lightbox images.

Metric	Mean value	Standard deviation
E_t [cm]	75.83	324.35
E_r [cm]	[0.32, 0.17, 69.41]	[22.75, 15.89, 324.61]
e_r [–]	$[0.559, 0.526, 107.01] \times 10^{-3}$	$[3.32, 2.80, 43.73] \times 10^{-2}$
e_q [deg]	18.00	35.81
E_θ [deg]	[14.99, 11.62, 12.89]	[32.81, 23.57, 27.48]
e_{SLAB} [–]	0.43162	0.88479

The augmentations exploited in the training partially compensate for the domain gap, but there are still margins for improvements. The already noticed characteristics of the proposed architecture using synthetic images in terms of error distributions are more evident from the scores in Table 7. Namely, the inaccuracies in the estimated keypoints drive the relative position error about the z -axis towards higher values than the error components along the x and y axes. On the contrary, the rotation errors are almost uniform for the three axes. Fig. 12 gives an overview of error trends and distributions. The image has been generated by following the same procedure outlined in Section 4.3.1 for the analogous plots (i.e., Fig. 8) and the shaded blue and gray areas refer to the 1σ and full SLAB error ranges, respectively. The normalized relative position and the relative attitude errors slightly increase as the ground truth distance increases, with a steeper increment of the error from about 7.5 m. Notably, there is a strong peak in the relative position error at about 8.5 m caused by a completely wrong estimate registered for a frame where the target is fully shadowed and not visible. Further, it has been noticed that the worst estimates are retrieved from images characterized by a target in low-illumination conditions, i.e., it is almost totally in shadow and poorly distinguishable from the background. Overall, the SLAB error distribution highlights outliers distributed through the entire dataset since the width of the shaded areas is almost constant for all the ground truth relative distances, except for the furthest images due to the previously mentioned increment of the error components.

Further analyzing the cases of worst estimates, it has been noticed that they are driven by a wrong ROI detection during the first inference that crops out almost entirely the darkest regions of the spacecraft in low contrast conditions with the background (see Fig. 13 where the ROI estimated from the YOLOv8s-pose model is reported), leading to a failure of all the subsequent steps. The same behavior holds for all the images that achieved SLAB scores higher than 3.0 (i.e., ~2% of the dataset). Notably, the case of a target with an almost null contrast with the background could not be compensated during the training phase via augmentations, suggesting that other strategies (e.g., sensor fusion as in [20]) could offer better results.

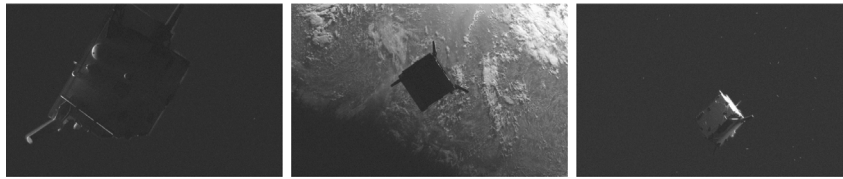


Fig. 11. Examples of SPEED+ Lightbox images.

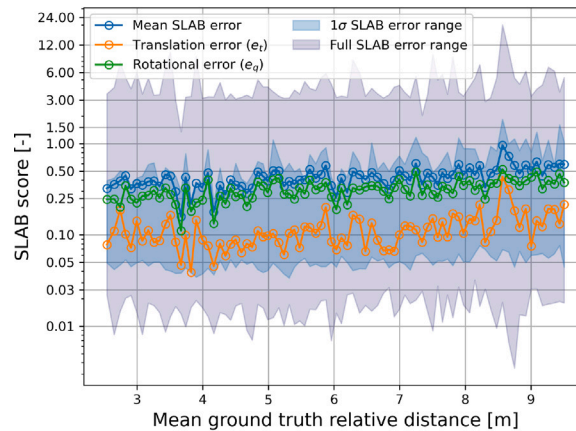


Fig. 12. Effects of relative distance on the SLAB score for SPEED+ Lightbox images.

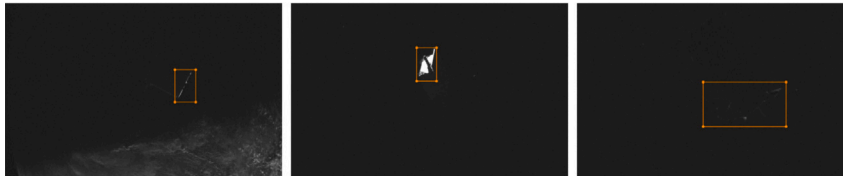


Fig. 13. Wrong ROI detection in low illumination conditions in SPEED+ Lightbox images.

SPEED+ sunlamp images. SPEED+ Sunlamp images include the effect of direct Sun illumination, leading to visible flares, even not in correspondence with the target due to the presence of lamps within the FOV, as shown in Fig. 14. Table 8 summarizes the performances of the YOLOv8s-pose architecture within this dataset. Even in that case, the performances of the proposed architecture are worse than the ones achieved on synthetic images, but they are comparable with those achieved on mock-up images belonging to the SPEED+ Lightbox dataset. Notably, the accuracy for Sunlamp images improves, and the uncertainties are lower than in the case of Lightbox images, pointing out the presence of fewer outliers. The metrics in Table 8 point out values similar to those already achieved on Lightbox images, i.e., higher uncertainties on the z-axis component of the translation error and a slightly uniform distribution of the angular errors on the three axes. The reduced amount of outliers and the consequent reduction of uncertainties can also be noticed in Fig. 15 that shows the mean SLAB score and its components (i.e., e_t and e_q) for increasing values of the ground truth distance (analogously to Fig. 12). The average relative attitude and position errors are almost constant throughout the dataset. The only peak in relative attitude and position errors is at about 8 m of relative distance. The uncertainties on the SLAB score are almost equally distributed throughout the entire dataset, as highlighted by the gray-shaded region that does not point out outliers as strong as for the case of SPEED+ Lightbox images. An increment of the errors, with a reduction of the dispersion of the SLAB error scores, is noticed for relative distances below 4 m, where the target is close to the camera. The high SLAB score is due to several failures, e.g., correctly retrieved ROI but outliers keypoint with high confidence associated and consequently not

Table 8

YOLOv8s-pose error metrics scores on SPEED+ Sunlamp images.

Metric	Mean value	Standard deviation
E_t [cm]	51.77	143.98
E_r [cm]	[0.24, 0.14, 34.23]	[19.73, 16.77, 146.86]
e_t [-]	$[0.181, 0.003, 57.27] \times 10^{-3}$	$[2.91, 2.40, 26.74] \times 10^{-2}$
e_q [deg]	19.80	36.22
E_θ [deg]	[16.63, 14.44, 12.27]	[33.41, 28.22, 25.28]
e_{SLAB} [-]	0.43183	0.77211

filtered by the algorithm, correct ROI estimation but completely wrong keypoint coordinates, and even failures in detecting the ROI. Examples of these failures are shown in Fig. 16, including the estimated ROI and keypoints, as well as the landmarks accuracy mapped in the color of the marker. These failures may arise from the fact that the YOLOv8s-pose model has not been directly trained from scratch on synthetic SPEED+ images, resulting in a model with better performances on a wide relative distance range but not optimized for the extremely low distances used to generate Sunlamp images. Hence, re-training from scratch a YOLOv8s-pose model only on SPEED+ synthetic images using the augmentations adopted for model M4 can be beneficial in increasing the overall performances on the mock-up image datasets at the cost of lower generalization capabilities to higher relative distance ranges. Despite that, it has been noticed that the algorithm is affected neither by the presence of lamps within the FOV nor by wrinkles in the multi-layer insulator material covering the mock-up and not simulated during the generation of the synthetic frames.

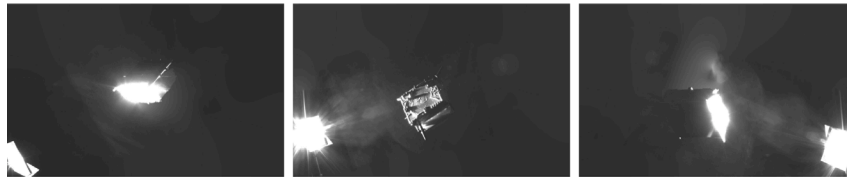


Fig. 14. Examples of SPEED+ Sunlamp images.

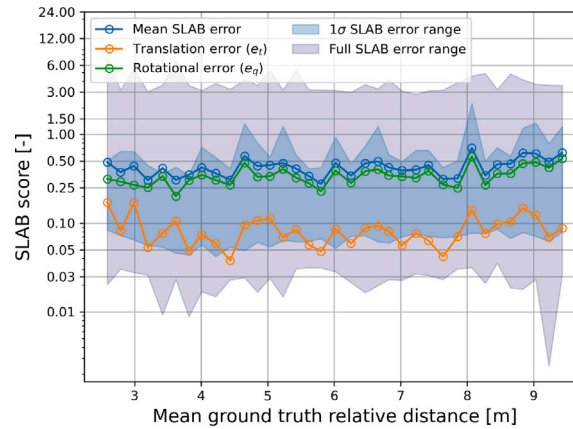


Fig. 15. Effects of relative distance on the SLAB score for SPEED+ Sunlamp images.

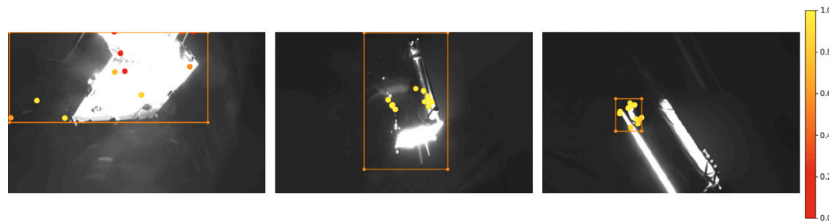


Fig. 16. Examples of failures of YOLOv8s-pose on SPEED+ Sunlamp images. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Comparison with other methods. Despite the performances on mock-up images being worse than those achieved on synthetic SPEED images due to the domain gap between the training and the testing sets, the proposed YOLOv8s-based architecture still achieves performances ranked within the top-10 best architectures among those that participated in the SPEC2021. A comparison of the proposed method with the outcomes of the SPEC2021 is given in Table 9, where the architectures are compared only by using the mean errors since the standard deviations are not available in literature [27]. Notice that the scores achieved by the PVSPE architecture available in [61] have been included in Table 9 for the sake of completeness despite not being included in the official leaderboards available on the SPEC2021 website. The YOLOv8s-pose-based architecture achieves the highest translational error on Lightbox images among the architectures in Table 9 due to the strong outlier discussed above given by a wrong ROI estimation. Instead, the angular error is comparable with the architectures that achieved similar SLAB scores since the quaternion error gives the highest contribution to the overall SLAB score. The performances on the Sunlamp images, in terms of both rotation and translation errors, are comparable with those of the architectures that achieved analogous SLAB scores. Notably, the architecture proposed here performs almost equally on the two datasets, highlighting that despite the Lightbox and Sunlamp images being very different, the YOLOv8s-pose bridges the domain gap among them. Unfortunately, it is impossible to compare the number of parameters of the proposed solution with all the models listed in Table 9 due to a lack of literature related to most of them. The only models in Table 9

for which the numbers of parameters are available are *TangoUnchained*, *lava1302*, *SPNv2*, and *PVSPE*. Namely, *TangoUnchained* adopted an architecture similar to the *UniAdelaide* model (tested on the SPEED dataset and proposed by the same authors of *TangoUnchained* [27]), consisting of $\sim 49.8\text{M}$ parameters. Notably, *TangoUnchained* leveraged adversarial training to achieve a low SLAB score on SPEED+ mock-up images. Concerning *lava1302*, it is disclosed in [27] that it consists of about 8.21M parameters. Despite that, it is worth pointing out that the low score achieved on the test mock-up images from SPEED+ is mostly due to the self-training of the architecture performed by relying on the test images themselves [27]. Instead, *SPNv2* consists of $\sim 52.5\text{M}$ parameters [29] and, also in this case, it shall be mentioned that the score registered on the SPEED+ has been obtained by performing online refinement on the test set [31,61]. Lastly, the *PVSPE* architecture consists of about 70M parameters [61]. It is worth pointing out that all these models rely on a higher amount of parameters than the solution proposed here, except for the case of *lava1302*. Despite that, the *PVSPE* has been successfully tested on an NVIDIA Jetson Xavier NX, achieving near real-time performances [61].

4.4. Extension to more complex geometries

Envisat was selected as the target model to prove the capabilities of the proposed architecture in more complex scenarios. Namely, Envisat is characterized by a shape more complex than Tango due to the

Table 9

Comparison of YOLOv8s-pose-based architecture against other methods on SPEED+ mock-up images.

Architecture	SPEED+ Lightbox			Architecture	SPEED+ Sunlamp		
	E_t [cm]	e_q [deg]	e_{SLAB} [–]		E_t [cm]	e_q [deg]	e_{SLAB} [–]
TangoUnchained	10.5	3.19	0.073	lava1302	6.5	2.73	0.059
PVSPE	0.017 ^a	4.81	0.101	VPU	7.6	2.83	0.061
VPU	13.1	4.58	0.101	TangoUnchained	8.6	4.30	0.090
SPNv2	15.0	5.58	0.122	u3s_lab	18.1	6.24	0.141
lava1302	30.2	6.66	0.165	haoranhuang_njust	16.3	8.41	0.175
haoranhuang_njust	18.0	8.13	0.173	PVSPE	0.022 ^a	8.94	0.178
u3s_lab	33.3	9.69	0.224	SPNv2	16.1	9.79	0.197
chusunhao	20.5	16.38	0.319	YOLOv8s-pose (our)	51.8	19.80	0.432
YOLOv8s-pose (our)	75.8	18.00	0.432	bbnc	54.2	21.95	0.465
for graduate	40.2	23.92	0.488	for graduate	45.5	22.97	0.487

^a Only the normalized adimensional translational error e_t is available from PVSPE architecture in [61].

abundance of appendages on the main body that cause frequent self-occlusions of the keypoints. Due to this, it has been decided to adopt 55 landmarks localized as in [78] on the main appendages of Envisat to ensure the visibility of most of them from each observation point. Fig. 17 shows the ground truth keypoints reprojected on a few images belonging to the Envisat dataset developed for the work in [78]. The landmarks and their visibility were already annotated in the dataset, while the ROI was labeled leveraging the already available keypoints and relative pose annotations. The images in Fig. 17 also show examples of labeled ROI. Notably, the keypoints belonging to the main solar array are involved only in the ROI definition, but they are not included in the keypoint annotations since they bring almost null improvements as they are out of the FOV for most images. The dataset was rendered using a Blender-based tool adopting the same camera model of MINIMA images [70] and it comprises 60 000 grayscale noisy images with target-to-camera relative distances in the range [5, 45] m. The images in the dataset have been shuffled and split into 45 000 training images, 9000 validation images, and 6000 testing images. The YOLOv8s-pose model has been trained on Envisat images for 900 epochs using the same parameters adopted for the training phase on SPEED images, including the baseline and VIS sensor noise augmentations. Notably, the model size increased to 13.2M parameters and 37.3 GFLOPs since it is requested to regress 55 keypoints. A tradeoff between the number of keypoints and the performances (i.e., accuracy and inference time) should be performed leveraging spaceborne hardware as possible future development. The trained model reached a mean AP_{50}^{95} of 99.4% and 93.9% in object detection and keypoint regression, respectively, with a mean IoU of 98.3% and a mean OKS of 96.4% on test images, pointing out a successful training phase and a high level of accuracy for both the tasks also applied to a target more complex than Tango.

The proposed YOLOv8s-pose-based architecture leveraged for SPEED images has been modified slightly due to the new target geometry. Namely, the checks on the need for a second inference have been simplified by removing the test on the aspect ratio of the ROI and the partial overlapping of the bounding box sides with the image borders since the size of Envisat and the relative distances at which the images are acquired make at least one of the bounding box sides to partially overlap with the image borders for almost all the frames in the dataset.

Additionally, the relative distance retrieval from the ROI and keypoints detected is avoided since it leads to highly inaccurate estimates due to the elongated shape of Envisat and since the target is partially out of the FOV in most of the images. Consequently, even the comparison step between the coarse position estimate and the one given as output by the EPnP is skipped.

Table 10 reports the error metrics computed for the Envisat test images. The low overall SLAB score achieved by the architecture highlights its high accuracy even for complex targets, proving the feasibility of the adopted approach for different geometries with a high number of features and a high probability of keypoints self-occlusions. Further, the post-processing of the YOLOv8s-pose outputs to retrieve the relative pose can be adapted easily to a wide range of geometries with only minor modifications. Notably, the mean attitude error is extremely low due to the high number of keypoints regressed by the YOLOv8s-pose model and their accuracy, with errors and uncertainties uniformly distributed on the three axes. It is worth mentioning that the keypoints distribution for Envisat leads to a higher keypoint inertia [21] about the target y-axis (normal to the long side of the main body and parallel to the Advanced Synthetic Aperture Radar). The mean translational error is in the order of approximately 10 cm, while its components show a higher error and uncertainties along the z-axis. Fig. 18 shows the trend of the SLAB score and its components through the dataset. The plot is generated by sorting the test images in ascending order of ground truth distance and generating batches of 300 images. The scores in the plot are the mean values in each batch, while the blue and gray shaded areas give the 1σ and the full SLAB error ranges, respectively. Notably, all the error metrics decrease as the distance increases due to the size of Envisat and the camera model adopted that, for low distances, led to a high amount of images with the target partially out of the FOV. This characteristic of the dataset also increases the uncertainties of the estimated relative pose, as shown by the widening of the gray shaded area and the shrinking of the blue one for distances below 10 m. Notably, the normalized relative translation error increases steeply for images generated with a relative distance below ~18 m, but the highest normalized mean relative translation error is still below 2.5%. Overall, the absence of error peaks in Fig. 18 excludes the presence of severe outliers, leading to SLAB scores below 0.05 for about 98% of the dataset

**Fig. 17.** Examples Envisat synthetic images with labeled ROI and keypoints (visible and non-visible).

Table 10
YOLOv8s-pose error metrics scores on Envisat images.

Metric	Mean value	Standard deviation
E_t [cm]	11.06	11.69
E_r [cm]	[−0.47, −0.10, −1.65]	[2.36, 2.25, 15.66]
e_t [−]	[−0.21, −0.008, −1.27] $\times 10^{-3}$	[0.23, 0.21, 1.20] $\times 10^{-2}$
e_q [deg]	0.64	1.13
E_θ [deg]	[0.47, 0.56, 0.47]	[0.88, 1.02, 0.89]
e_{SLAB} [−]	0.01651	0.02771

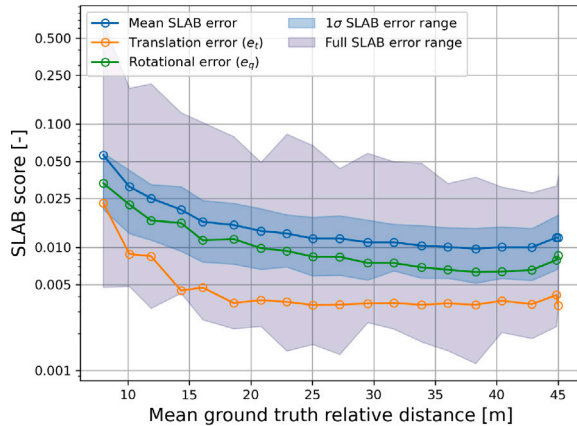


Fig. 18. Effects of relative distance on the SLAB score for Envisat images. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

and a maximum SLAB error of about 0.67. Notably, all the images with an associated slab score higher than 0.1 are characterized by low relative distances and adverse illumination conditions, with only a few portions of the appendages distinguishable from the background. Lastly, despite the higher number of parameters and GFLOPs than the case of Tango as a target due to the 55 keypoints requested as output, the overall pipeline achieved a mean running time of about 80 ms for the case of single inference, and about 165 ms when two inferences are performed.

To further assess the effectiveness of the proposed approach for more complex targets, it has been attempted to deal with images using VESPA as a target, i.e., a highly symmetric space-resident object with only small appendages on the external surface that cannot be easily detected, as well as a central part of the main body made of carbon-fiber composite that has low contrast with the deep space background. A synthetic dataset of 33 000 RGB annotated images using VESPA as a target has been generated resembling the distribution of MINIMA [70], using relative distances within the range [10, 100] m. Fig. 19 shows examples of images from the VESPA dataset.

The training was performed by testing several combinations of hyperparameters and keypoint configurations, but despite that, the CNN is not capable of retrieving the correct keypoint locations. Namely,

the model learns how to correctly perform the target detection task in a few training steps, while the metrics associated with the keypoints regression accuracy increase slowly and saturate early, reaching the plateaux at a low level of accuracy. These outcomes point out that the YOLOv8s-pose model cannot regress the location of keypoints for highly symmetric targets despite being capable of accurately estimating the ROI also for symmetric targets, pointing out an applicability limit for the proposed architecture

5. Conclusions and future works

Autonomous spacecraft relative navigation is an active field of research due to the attractive and challenging missions that it can enable. The development of relative pose estimation algorithms from monocular images for known uncooperative targets received a boost in recent years due to the introduction of AI-based solutions and the publication of benchmark datasets. Despite that, the results of the satellites pose estimation challenges sponsored by ESA point out that the best-performing architectures have been developed by focusing primarily on enhancing accuracy, disregarding the computational constraints due to the currently available space-grade hardware. Further, making the architectures capable of generalizing well from the synthetic images used within the training phase to actual mock-up frames representative of the actual mission scenario is still a challenge. Both these issues are addressed in this work by proposing a novel, robust, and efficient relative pose initialization algorithm based on a single multitasking CNN. Namely, the proposed architecture leverages the YOLOv8s-pose to perform ROI estimation and keypoint regression in a single inference. The proposed pipeline, leveraging the single CNN architecture, features an increased robustness due to the implemented outlier rejection scheme that relies on the confidence score associated with each keypoint and given as output by the YOLOv8s-pose model. Further, the efficiency is enhanced by leveraging a novel scheme that evaluates the need for a second inference, making it possible to rely on the output of the first inference (i.e., without performing the cropping of the image to the retrieved ROI) to estimate the relative pose directly. Notably, this feature of the proposed architecture may enable the possibility of increasing the update rate of the navigation filters for the closest approach. Remarkably, relying on a CNN that outputs the retrieved landmarks in a fixed order increases the efficiency of the pipeline since it requires the solution of only a single PnP problem as the 2D-to-3D features matches are already known. The YOLOv8s-pose model has been trained by adopting custom synthetic augmentations to reduce the domain gap between synthetic and mock-up images. Notably, a novel augmentation that applies a random level of noise with a high degree of representativeness of actual VIS sensor noises to the processed training images is introduced within this work and proven to be highly beneficial in enhancing both the target detection and keypoint regression performances of the CNN both in synthetic SPEED images and mock-up SPEED+ frames. Further, the YOLOv8s-pose, trained within this work adopting the discussed augmentation strategies, is the best-performing model on synthetic speed images in both target detection and keypoint

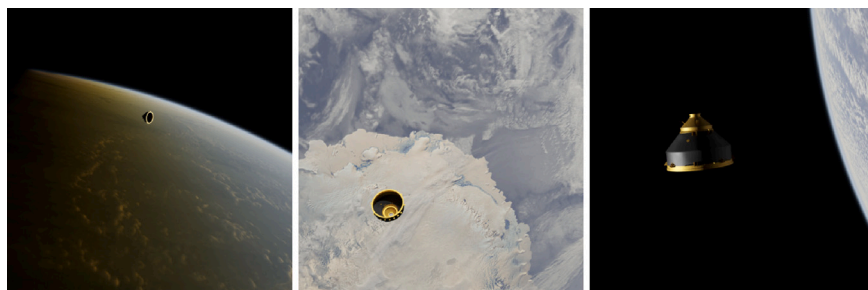


Fig. 19. Examples of synthetic images extracted from the VESPA dataset.

regression tasks, outperforming other models available in the literature. The proposed architecture achieved the third-best score on synthetic SPEED images, corresponding to a mean SLAB score of 0.03305, i.e., a mean translational error of about 9.5 cm and a mean attitude error of 1.4 degrees. Further, the analyses of the errors point out that both the accuracy and precision are high due to the absence of outliers, resulting in the most precise model in relative attitude estimation among those that participated in SPEC2019, despite being also one of the lightest architectures. Additionally, even the computational time measured on a CPU is well promising, being about 60 ms for a single inference and about 131 ms for the cases in which the relative pose retrieval needs two inferences. The domain gap bridging capabilities have been assessed by processing mock-up images from the SPEED+ dataset. The outcomes of the analyses revealed that the overall performances are still within the top 10 registered during SPEC2021, despite the errors increased with respect to synthetic images, proving that the augmentations introduced are effective in reducing the domain gap, resulting in a lightweight architecture that does not need a discriminator module as the best-performing ones. The analyses of the mock-up frames associated with the worst estimates pointed out that the highest errors are due to extremely low illumination conditions or wrong detections (of the target or the keypoints). In the former case, possible mitigations can involve exploiting sensors operating in different spectral bands since dedicated augmentations may be ineffective. Conversely, mitigating the issues related to the latter case can involve training only on synthetic SPEED+ images, i.e., reducing the relative range gap between SPEED and SPEED+ images. The architecture has also been tested on datasets using Envisat and VESPA as targets to prove the robustness of the proposed approach with respect to more complex targets. In the former case, the algorithm has been slightly modified, relaxing the checks on the need for a second inference and removing the validity check on the retrieved relative position. Despite that, the performances are still impressive both for the accuracy of the retrieved pose (i.e., 11 cm of mean relative position error and about 0.6 degrees of relative attitude error) and overall computational time on CPU that is at most 165 ms despite the CNN is requested to regress 55 keypoints. Unfortunately, the same performances have not been achieved for VESPA, i.e., a highly symmetric target, for which the training was not successful independently for the keypoints selected, resulting in a CNN that is capable of accurately retrieving only the ROI but not the keypoints.

This behavior for highly symmetric targets poses an applicability limit for the proposed architecture that shall be further investigated in future developments aimed at identifying the best strategies to deal with that or, eventually, relying on a simplified version that estimates the relative position and the direction of the axis of symmetry instead of a complete relative pose. Further, despite the well promising results related to the CPU computational time, testing the proposed architecture on space-grade hardware is still needed to ensure onboard deployability. Lastly, the analysis of the outputs of the proposed algorithms revealed that the worst estimates are achieved mainly in low illumination conditions. Hence, solutions that also involve sensors operating in other spectral bands (e.g., the thermal infrared range) should be evaluated to assess possible improvements in the reliability and robustness of the proposed solution.

CRedit authorship contribution statement

Michele Bechini: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Michèle Lavagna:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

References

- [1] J.A. Starek, B. Açıkmeşe, I.A. Nesnas, M. Pavone, Spacecraft autonomy challenges for next-generation space missions, in: E. Feron (Ed.), *Advances in Control System Technology for Aerospace Applications*, Springer Berlin Heidelberg, Berlin, Heidelberg, 2016, pp. 1–48, http://dx.doi.org/10.1007/978-3-662-47694-9_1.
- [2] L. Zhang, S. Zhang, H. Yang, H. Cai, S. Qian, Relative attitude and position estimation for a tumbling spacecraft, *Aerosp. Sci. Technol.* 42 (2015) 97–105, <http://dx.doi.org/10.1016/j.ast.2014.12.025>.
- [3] Q. Li, J. Yuan, B. Zhang, C. Gao, Model predictive control for autonomous rendezvous and docking with a tumbling target, *Aerosp. Sci. Technol.* 69 (2017) 700–711, <http://dx.doi.org/10.1016/j.ast.2017.07.022>.
- [4] R. Biesbroek, S. Aziz, A. Wolahan, S. Cipolla, M. Richard-Noca, L. Piguet, The clearspace-1 mission: Esa and clearspace team up to remove debris, in: *Proc. 8th Eur. Conf. Sp. Debris*, 2021, pp. 1–3.
- [5] L. Sun, W. Huo, Robust adaptive relative position tracking and attitude synchronization for spacecraft rendezvous, *Aerosp. Sci. Technol.* 41 (2015) 28–35, <http://dx.doi.org/10.1016/j.ast.2014.11.013>.
- [6] L. Breger, J.P. How, Safe trajectories for autonomous rendezvous of spacecraft, *J. Guid. Control Dyn.* 31 (5) (2008) 1478–1489, <http://dx.doi.org/10.2514/1.29590>.
- [7] S. Li, P. Cui, H. Cui, Vision-aided inertial navigation for pinpoint planetary landing, *Aerosp. Sci. Technol.* 11 (6) (2007) 499–506, <http://dx.doi.org/10.1016/j.ast.2007.04.006>.
- [8] P. Lunghi, M. Ciarambino, M. Lavagna, A multilayer perceptron hazard detector for vision-based autonomous planetary landing, *Adv. Space Res.* 58 (1) (2016) 131–144, <http://dx.doi.org/10.1016/j.asr.2016.04.012>.
- [9] S. Silvestrini, M. Piccinin, G. Zanotti, A. Brandonisio, I. Bloise, L. Feruglio, P. Lunghi, M. Lavagna, M. Varile, Optical navigation for lunar landing based on convolutional neural network crater detector, *Aerosp. Sci. Technol.* 123 (2022) 107503, <http://dx.doi.org/10.1016/j.ast.2022.107503>.
- [10] S. Li, P. Cui, H. Cui, Autonomous navigation and guidance for landing on asteroids, *Aerosp. Sci. Technol.* 10 (3) (2006) 239–247, <http://dx.doi.org/10.1016/j.ast.2005.12.003>.
- [11] P. Tian, M. Yao, X. Xiao, B. Zheng, T. Cao, Y. Xi, H. Liu, H. Cui, 3-D semantic terrain reconstruction of monocular close-up images of martian terrains, *IEEE Trans. Geosci. Remote Sens.* 62 (2024) 1–16, <http://dx.doi.org/10.1109/TGRS.2024.3378192>.
- [12] G. Di Mauro, M. Lawn, R. Bevilacqua, Survey on guidance navigation and control requirements for spacecraft formation-flying missions, *J. Guid. Control Dyn.* 41 (3) (2018) 581–602, <http://dx.doi.org/10.2514/1.G002868>.
- [13] S. Silvestrini, M. Lavagna, Neural-based predictive control for safe autonomous spacecraft relative maneuvers, *J. Guid. Control Dyn.* 44 (12) (2021) 2303–2310, <http://dx.doi.org/10.2514/1.G005481>.
- [14] L. Pasqualetto Cassinis, R. Fonod, E. Gill, Review of the robustness and applicability of monocular pose estimation systems for relative navigation with an uncooperative spacecraft, *Prog. Aerosp. Sci.* 110 (2019) 100548, <http://dx.doi.org/10.1016/j.paerosci.2019.05.008>.
- [15] V. Pesce, M.F. Haydar, M. Lavagna, M. Lovera, Comparison of filtering techniques for relative attitude estimation of uncooperative space objects, *Aerosp. Sci. Technol.* 84 (2019) 318–328, <http://dx.doi.org/10.1016/j.ast.2018.10.031>.
- [16] V. Pesce, S. Silvestrini, M. Lavagna, Radial basis function neural network aided adaptive extended Kalman filter for spacecraft relative navigation, *Aerosp. Sci. Technol.* 96 (2020) 105527, <http://dx.doi.org/10.1016/j.ast.2019.105527>.
- [17] R. Opromolla, G. Fasano, G. Rufino, M. Grassi, A review of cooperative and uncooperative spacecraft pose determination techniques for close-proximity operations, *Prog. Aerosp. Sci.* 93 (2017) 53–72, <http://dx.doi.org/10.1016/j.paerosci.2017.07.001>.
- [18] V. Pesce, R. Opromolla, S. Sarno, M. Lavagna, M. Grassi, Autonomous relative navigation around uncooperative spacecraft based on a single camera, *Aerosp. Sci. Technol.* 84 (2019) 1070–1080, <http://dx.doi.org/10.1016/j.ast.2018.11.042>.
- [19] T. Jiang, C. Yunfeng, D. Meng, Z. Zhouyu, Visible and infrared image fusion-based image quality enhancement with applications to space debris on-orbit surveillance, *Int. J. Aerosp. Eng.* 2022 (1) (2022) 1–21, <http://dx.doi.org/10.1155/2022/6300437>.
- [20] G.L. Civardi, M. Bechini, M. Quirino, A. Colombo, M. Piccinin, M. Lavagna, Generation of fused visible and thermal-infrared images for uncooperative spacecraft proximity navigation, *Adv. Space Res.* (2023) <http://dx.doi.org/10.1016/j.asr.2023.03.022>.
- [21] M. Piazza, M. Maestrini, P. Di Lizia, Monocular relative pose estimation pipeline for uncooperative resident space objects, *J. Aerosp. Inf. Syst.* 19 (9) (2022) 613–632, <http://dx.doi.org/10.2514/1.1011064>.

- [22] B. Chen, J. Cao, A. Parra, T.-J. Chin, Satellite pose estimation with deep landmark regression and nonlinear pose refinement, in: *Proceedings of the IEEE/CVF International Conference on Computer Vision Workshops*, 2019.
- [23] M. Kisantal, S. Sharma, T.H. Park, D. Izzo, M. Märtens, S. D'Amico, Spacecraft pose estimation dataset (SPEED), Zenodo (2019) <http://dx.doi.org/10.5281/zenodo.6327546>.
- [24] T.H. Park, M. Märtens, G. Lecuyer, D. Izzo, S. D'Amico, Next generation spacecraft pose estimation dataset (SPEED+), Zenodo (2021) <http://dx.doi.org/10.25740/wv398fc4383>.
- [25] S. D'Amico, P. Bodin, M. Delpach, R. Noteborn, PRISMA, in: M. D'Errico (Ed.), *Distributed Space Missions for Earth System Monitoring*, Springer Science & Business Media, 2013, pp. 599–637.
- [26] M. Kisantal, S. Sharma, T.H. Park, D. Izzo, M. Märtens, S. D'Amico, Satellite pose estimation challenge: Dataset, competition design, and results, *IEEE Trans. Aerosp. Electron. Syst.* 56 (5) (2020) 4083–4098, <http://dx.doi.org/10.1109/TAES.2020.2989063>.
- [27] T.H. Park, M. Märtens, M. Jawaid, Z. Wang, B. Chen, T.-J. Chin, D. Izzo, S. D'Amico, Satellite pose estimation competition 2021: Results and analyses, *Acta Astronaut.* (2023) <http://dx.doi.org/10.1016/j.actaastro.2023.01.002>.
- [28] V. Lepetit, F. Moreno-Noguer, P. Fua, EPnP: An accurate o(n) solution to the PnP problem, *Int. J. Comput. Vis.* 81 (2) (2009) 155–166, <http://dx.doi.org/10.1007/s11263-008-0152-6>.
- [29] L. Pauly, W. Rharbaoui, C. Shneider, A. Rathinam, V. Gaudillière, D. Aouada, A survey on deep learning-based monocular spacecraft pose estimation: Current state, limitations and prospects, *Acta Astronaut.* 212 (2023) 339–360, <http://dx.doi.org/10.1016/j.actaastro.2023.08.001>, URL <https://www.sciencedirect.com/science/article/pii/S0094576523003995>.
- [30] D. Kaidanovic, *AI-Aided Optical Navigation about Uncooperative Spacecraft Using Synthetic Imagery Algorithm for Non-Cooperative Spacecrafts* (MSc Thesis), Politecnico di Milano, 2022.
- [31] T.H. Park, S. D'Amico, Robust multi-task learning and online refinement for spacecraft pose estimation across domain gap, *Adv. Space Res.* (2023) <http://dx.doi.org/10.1016/j.asr.2023.03.036>.
- [32] H. Liu, M. Yao, X. Xiao, B. Zheng, H. Cui, MarsScapes and UDAFormer: A panorama dataset and a transformer-based unsupervised domain adaptation framework for martian terrain segmentation, *IEEE Trans. Geosci. Remote Sens.* 62 (2024) 1–17, <http://dx.doi.org/10.1109/TGRS.2023.3343109>.
- [33] M. Bechini, G. Gu, P. Lunghi, M. Lavagna, Robust spacecraft relative pose estimation via CNN-aided line segments detection in monocular images, *Acta Astronaut.* 215 (2024) 20–43, <http://dx.doi.org/10.1016/j.actaastro.2023.11.049>.
- [34] S. Sharma, J. Ventura, S. D'Amico, Robust model-based monocular pose initialization for noncooperative spacecraft rendezvous, *AIAA J. Spacecr. Rocket.* 55 (6) (2018) 1414–1429, <http://dx.doi.org/10.2514/1.A34124>.
- [35] J. Terven, D. Cordova-Esparza, A comprehensive review of YOLO: From YOLOv1 to YOLOv8 and beyond, 2023, arXiv preprint [arXiv:2304.00501](https://arxiv.org/abs/2304.00501).
- [36] P. Mittraipyanumic, G. DeSouza, A. Kak, Calculating the 3d-pose of rigid-objects using active appearance models, in: *IEEE International Conference on Robotics and Automation*, 2004. *Proceedings. ICRA '04*, 2004, Vol. 5, 2004, pp. 5147–5152, <http://dx.doi.org/10.1109/ROBOT.2004.1302534>.
- [37] R. Opromolla, G. Fasano, G. Rufino, M. Grassi, A model-based 3D template matching technique for pose acquisition of an uncooperative space object, *Sensors* 15 (3) (2015) 6360–6382, <http://dx.doi.org/10.3390/s150306360>.
- [38] J.-F. Shi, S. Ulrich, S. Ruel, Spacecraft pose estimation using principal component analysis and a monocular camera, in: *AIAA Guidance, Navigation, and Control Conference*, 2017, pp. 1–24, <http://dx.doi.org/10.2514/6.2017-1034>.
- [39] S. Sharma, S. D'Amico, Comparative assessment of techniques for initial pose estimation using monocular vision, *Acta Astronaut.* 123 (2016) 435–445, <http://dx.doi.org/10.1016/j.actaastro.2015.12.032>.
- [40] R. Opromolla, C. Vela, A. Nocerino, C. Lombardi, Monocular-based pose estimation based on fiducial markers for space robotic capture operations in GEO, *Remote. Sens.* 14 (18) (2022) <http://dx.doi.org/10.3390/rs14184483>.
- [41] S. D'Amico, M. Benn, J.L. Jørgensen, Pose estimation of an uncooperative spacecraft from actual space imagery, *Int. J. Space Sci. Eng.* 2 (2) (2014) 171–189, <http://dx.doi.org/10.1504/IJSPACESE.2014.060600>.
- [42] B.J. Naasz, R.D. Bums, S.Z. Queen, J. Van Eepoel, J. Hannah, E. Skelton, The HST SM4 relative navigation sensor system: Overview and preliminary testing results from the flight robotics lab, *J. Astronaut. Sci.* 57 (1–2) (2009) 457–483, <http://dx.doi.org/10.1007/BF03321512>.
- [43] E. Karami, S. Prasad, M. Shehata, Image matching using SIFT, SURF, BRIEF and ORB: performance comparison for distorted images, 2017, arXiv preprint [arXiv:1710.02726](https://arxiv.org/abs/1710.02726).
- [44] S.A.K. Tareen, Z. Saleem, A comparative analysis of SIFT, SURF, KAZE, AKAZE, ORB, and BRISK, in: *2018 International Conference on Computing, Mathematics and Engineering Technologies (ICOMET)*, 2018, pp. 1–10, <http://dx.doi.org/10.1109/ICOMET.2018.8346440>.
- [45] D. Bojanić, K. Bartol, T. Pribanić, T. Petković, Y.D. Donoso, J.S. Mas, On the comparison of classic and deep keypoint detector and descriptor methods, in: *2019 11th International Symposium on Image and Signal Processing and Analysis, ISPA*, 2019, pp. 64–69, <http://dx.doi.org/10.1109/ISPA.2019.8868792>.
- [46] V. Capuano, S.R. Alimo, A.Q. Ho, S.-J. Chung, Robust features extraction for on-board monocular-based spacecraft pose acquisition, in: *AIAA Scitech 2019 Forum*, Vol. 1, 2019, pp. 1–15, <http://dx.doi.org/10.2514/6.2019-2005>.
- [47] T.H. Park, S. Sharma, S. D'Amico, Towards robust learning-based pose estimation of noncooperative spacecraft, 2019, arXiv preprint [arXiv:1909.00392](https://arxiv.org/abs/1909.00392).
- [48] S. Mahendran, H. Ali, R. Vidal, 3D pose regression using convolutional neural networks, in: *2017 IEEE Conference on Computer Vision and Pattern Recognition Workshops*, CVPRW, 2017, pp. 494–495, <http://dx.doi.org/10.1109/CVPRW.2017.73>.
- [49] P.F. Proença, Y. Gao, Deep learning for spacecraft pose estimation from photorealistic rendering, in: *2020 IEEE International Conference on Robotics and Automation, ICRA*, IEEE, 2020, pp. 6007–6013.
- [50] L. Pasqualetto Cassinis, R. Fonod, E. Gill, I. Ahms, J. Gil-Fernández, Evaluation of tightly- and loosely-coupled approaches in CNN-based pose estimation systems for uncooperative spacecraft, *Acta Astronaut.* 182 (2021) 189–202, <http://dx.doi.org/10.1016/j.actaastro.2021.01.035>.
- [51] M. Bechini, *Monocular Vision for Uncooperative Targets through AI-Based Methods and Sensors Fusion* (Ph.D. thesis), Politecnico di Milano, 2024, URL <https://hdl.handle.net/10589/216793>.
- [52] S. Ren, K. He, R. Girshick, J. Sun, Faster r-cnn: Towards real-time object detection with region proposal networks, *Adv. Neural Inf. Process. Syst.* 28 (2015).
- [53] K. Sun, B. Xiao, D. Liu, J. Wang, Deep high-resolution representation learning for human pose estimation, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2019, pp. 5693–5703.
- [54] Y. Hu, J. Hugonot, P. Fua, M. Salzmann, Segmentation-driven 6d object pose estimation, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2019, pp. 3385–3394.
- [55] G. Gu, B.S. Ko, S.H. Go, S. Lee, M. Shin, Towards real-time and lightweight line segment detection, 2021, CoRR abs/2106.00186, arXiv:2106.00186, URL <https://arxiv.org/abs/2106.00186>.
- [56] K. He, X. Zhang, S. Ren, J. Sun, Deep residual learning for image recognition, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2016, pp. 770–778.
- [57] Z. Wang, M. Chen, Y. Guo, Z. Li, Q. Yu, Bridging the domain gap in satellite pose estimation: A self-training approach based on geometrical constraints, *IEEE Trans. Aerosp. Electron. Syst.* (2023) 1–14, <http://dx.doi.org/10.1109/TAES.2023.3250385>.
- [58] Y. Yuan, R. Fu, L. Huang, W. Lin, C. Zhang, X. Chen, J. Wang, HRFormer: High-resolution vision transformer for dense predict, in: M. Ranzato, A. Beygelzimer, Y. Dauphin, P. Liang, J.W. Vaughan (Eds.), *Advances in Neural Information Processing Systems*, Vol. 34, Curran Associates, Inc., 2021, pp. 7281–7293.
- [59] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, H. Adam, Encoder-decoder with atrous separable convolution for semantic image segmentation, in: *Proceedings of the European Conference on Computer Vision, ECCV*, 2018, pp. 801–818.
- [60] M. Tan, R. Pang, Q.V. Le, EfficientDet: Scalable and efficient object detection, in: *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR*, 2020, pp. 10778–10787, <http://dx.doi.org/10.1109/CVPR42600.2020.01079>.
- [61] H. Yang, X. Xiao, M. Yao, Y. Xiong, H. Cui, Y. Fu, PVSPE: A pyramid vision multitask transformer network for spacecraft pose estimation, *Adv. Space Res.* 74 (3) (2024) 1327–1342, <http://dx.doi.org/10.1016/j.asr.2024.05.011>.
- [62] E. Samkari, M. Arif, M. Alghamdi, M.A. Al Ghamdi, Human pose estimation using deep learning: A systematic literature review, *Mach. Learn. Knowl. Extr.* 5 (4) (2023) 1612–1659, <http://dx.doi.org/10.3390/make5040081>.
- [63] D. Maji, S. Nagori, M. Mathew, D. Poddar, Yolo-pose: Enhancing yolo for multi person pose estimation using object keypoint similarity loss, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 2637–2646.
- [64] C.-Y. Wang, H.-Y.M. Liao, Y.-H. Wu, P.-Y. Chen, J.-W. Hsieh, I.-H. Yeh, CSPNet: A new backbone that can enhance learning capability of CNN, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, 2020, pp. 390–391.
- [65] K. He, X. Zhang, S. Ren, J. Sun, Spatial pyramid pooling in deep convolutional networks for visual recognition, *IEEE Trans. Pattern Anal. Mach. Intell.* 37 (9) (2015) 1904–1916.
- [66] T.-Y. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, S. Belongie, Feature pyramid networks for object detection, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2017, pp. 2117–2125.
- [67] S. Liu, L. Qi, H. Qin, J. Shi, J. Jia, Path aggregation network for instance segmentation, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2018, pp. 8759–8768.
- [68] R. Qiao, G. Xu, P. Wang, Y. Cheng, W. Dong, An accurate, efficient, and stable perspective-n-point algorithm in 3D space, *Appl. Sci.* 13 (2) (2023) 1111, <http://dx.doi.org/10.3390/app13021111>.
- [69] T. Möller, B. Trumbore, Fast, minimum storage ray-triangle intersection, *J. Graph. Tools* 2 (1) (1997) 21–28, <http://dx.doi.org/10.1080/10867651.1997.10487468>.
- [70] M. Bechini, M. Lavagna, P. Lunghi, Dataset generation and validation for spacecraft pose estimation via monocular images processing, *Acta Astronaut.* 204 (2023) 358–369, <http://dx.doi.org/10.1016/j.actaastro.2023.01.012>.

- [71] M. Konnik, J. Welsh, High-level numerical simulations of noise in CCD and CMOS photosensors: review and tutorial, 2014, arXiv preprint [arXiv:1412.4031](https://arxiv.org/abs/1412.4031).
- [72] K. Wei, Y. Fu, Y. Zheng, J. Yang, Physics-based noise modeling for extreme low-light photography, *IEEE Trans. Pattern Anal. Mach. Intell.* 44 (11) (2022) 8520–8537, [http://dx.doi.org/10.1109/TPAMI.2021.3103114](https://doi.org/10.1109/TPAMI.2021.3103114).
- [73] European Machine Vision Association, EMVA 1288 Standard: Standard for Characterization of Image Sensors and Cameras, Tech. Rep. Release 4.0 Linear, EMVA, 2021, URL https://www.emva.org/wp-content/uploads/EMVA1288Linear_4.0Release.pdf.
- [74] A. Buslaev, V.I. Iglovikov, E. Khvedchenya, A. Parinov, M. Druzhinin, A.A. Kalinin, Albumentations: Fast and flexible image augmentations, *Inf.* 11 (2) (2020) [http://dx.doi.org/10.3390/info11020125](https://doi.org/10.3390/info11020125).
- [75] P.T. Jackson, A. Atapour-Abarghouei, S. Bonner, T.P. Breckon, B. Obara, Style augmentation: Data augmentation via style randomization, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops*, 2019, pp. 83–92.
- [76] G. Ghiasi, H. Lee, M. Kudlur, V. Dumoulin, J. Shlens, Exploring the structure of a real-time, arbitrary neural artistic stylization network, in: *British Machine Vision Conference*, 2017, [http://dx.doi.org/10.48550/arXiv.1705.06830](https://doi.org/10.48550/arXiv.1705.06830).
- [77] S. Sharma, S. D'Amico, Neural network-based pose estimation for noncooperative spacecraft rendezvous, *IEEE Trans. Aerosp. Electron. Syst.* 56 (6) (2020) 4638–4658, [http://dx.doi.org/10.1109/TAES.2020.2999148](https://doi.org/10.1109/TAES.2020.2999148).
- [78] M. Quintana, B. Parreira, R. Hinz, J. Belfo, P. Rosa, J. Balsalobre, F. Membibre, A. Latorre, F. Buckley, J. Luis Espinosa-Aranda, et al., AIVIONIC-artificial intelligence techniques in on-board avionics and software, in: *12th International Conference on Guidance, Navigation & Control Systems (GNC) and 9th International Conference on Astrodynamics Tools and Techniques, ICATT*, 2023, pp. 1–15.